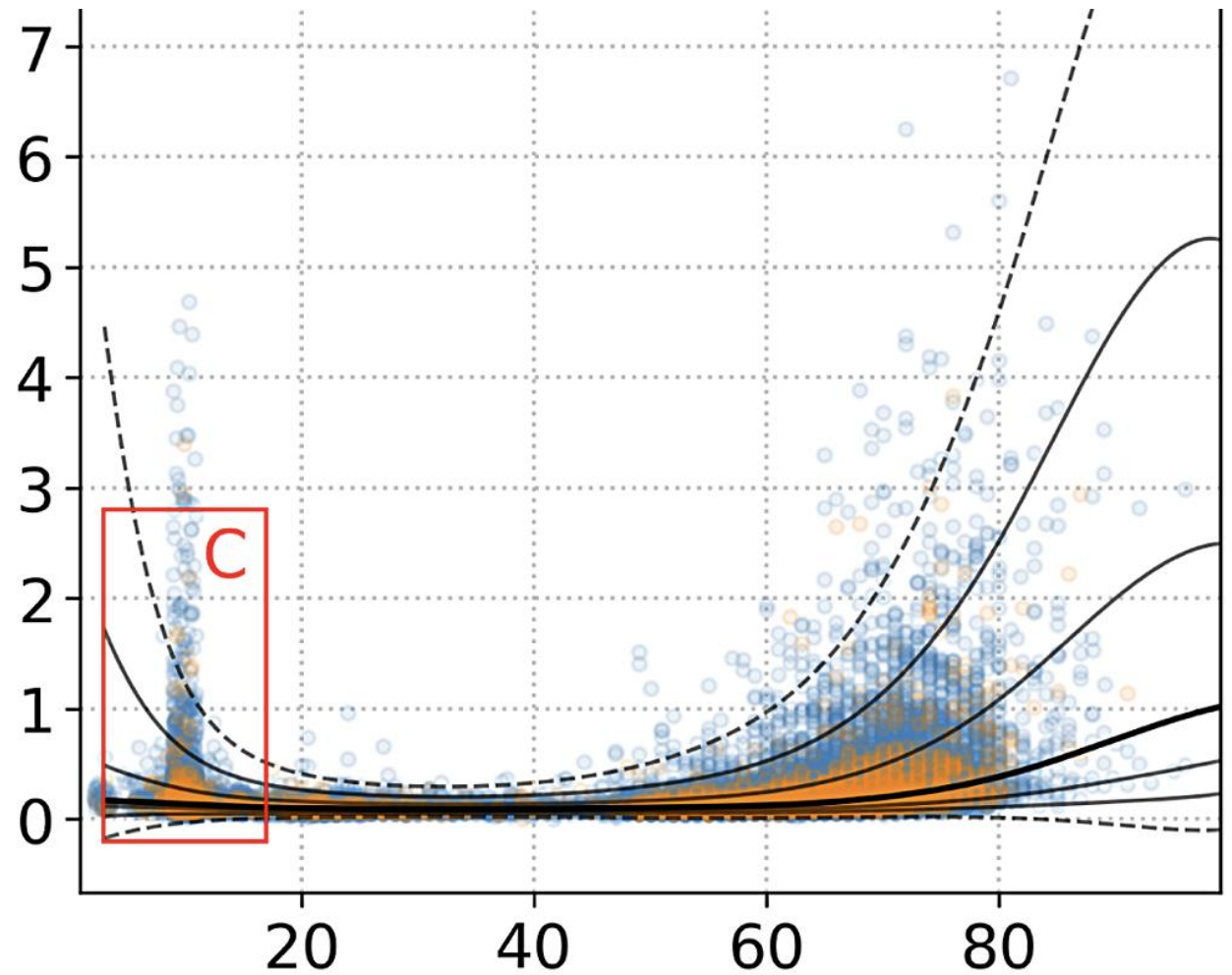


Normative modelling in the Life Sciences

Lecture by: Michelle Wang, Johanna Bayer



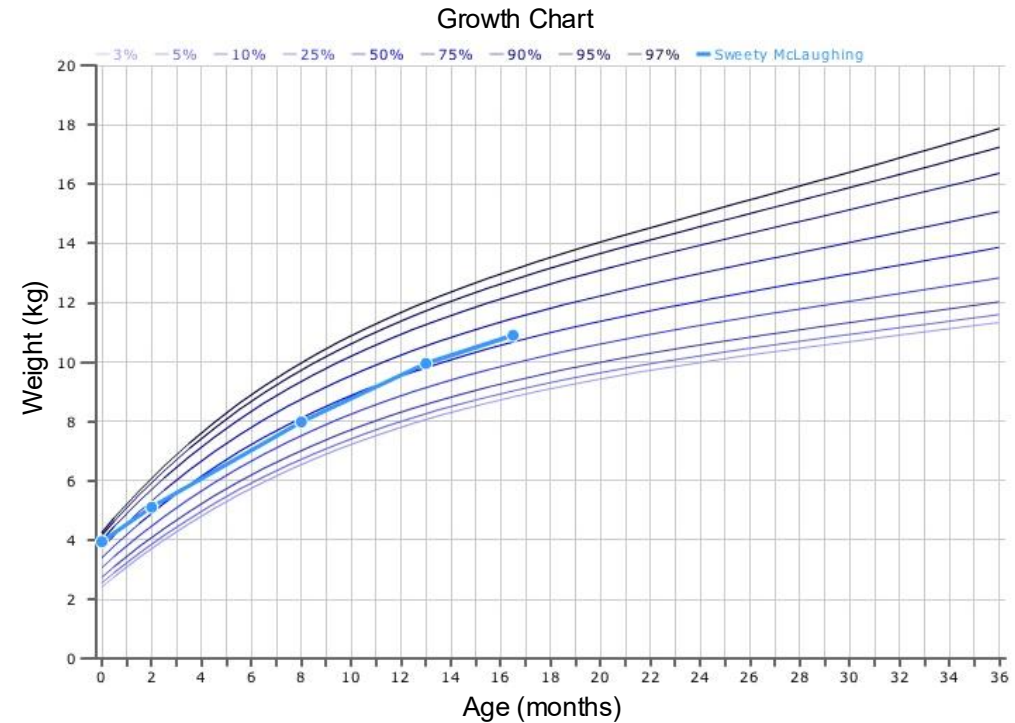
Outline

- Part 1: Theoretical Introduction to normative modelling
 - What are normative models and how are they created?
 - Why normative modelling in the life sciences?
 - Previous applications
 - The normative model using hierarchical Bayesian Regression
 - Notebook preparation

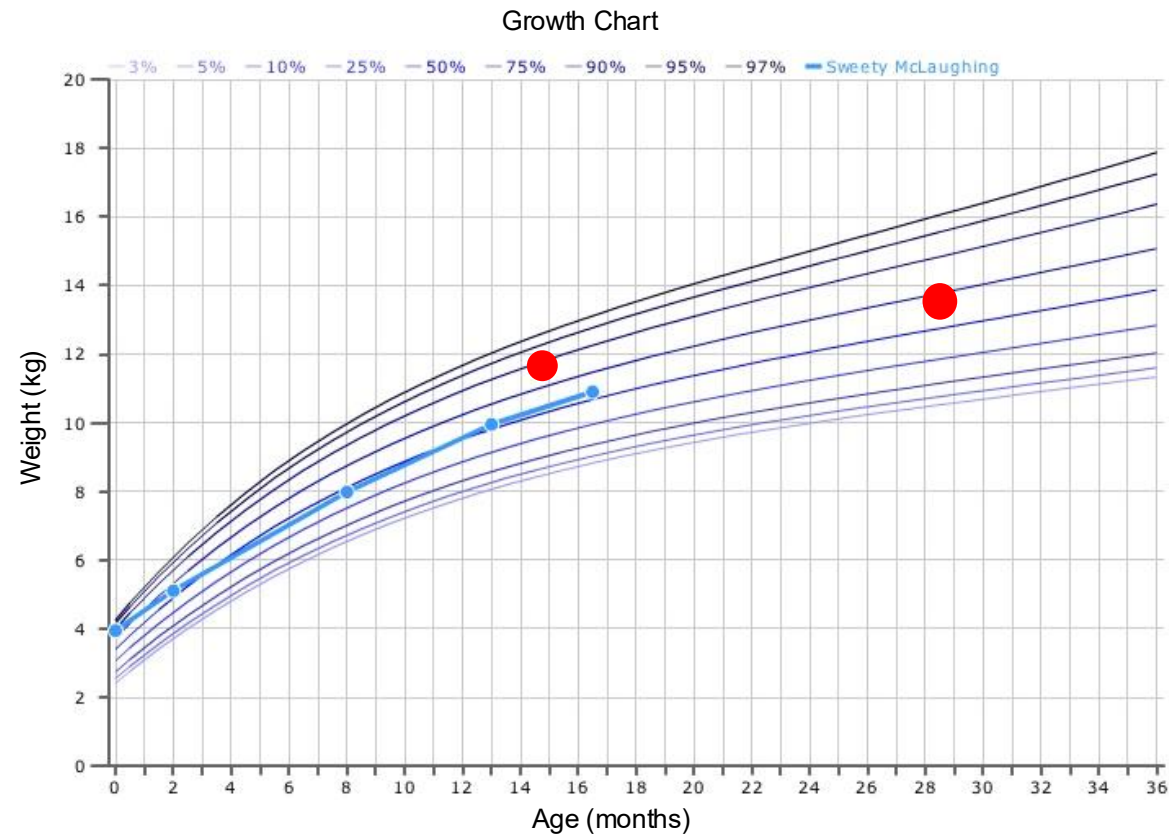
Part 2: Fitting various normative models

Growth Charts

- How can we compare children in their weight or height while they grow?
- Reference height and weight to a cohort of similar individuals

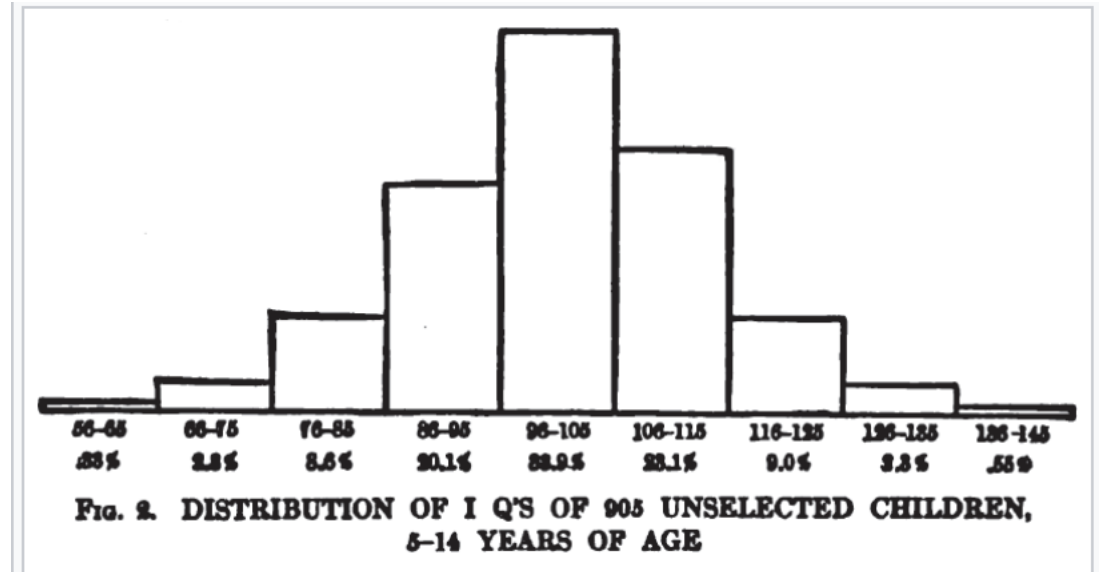


Growth Charts



IQ

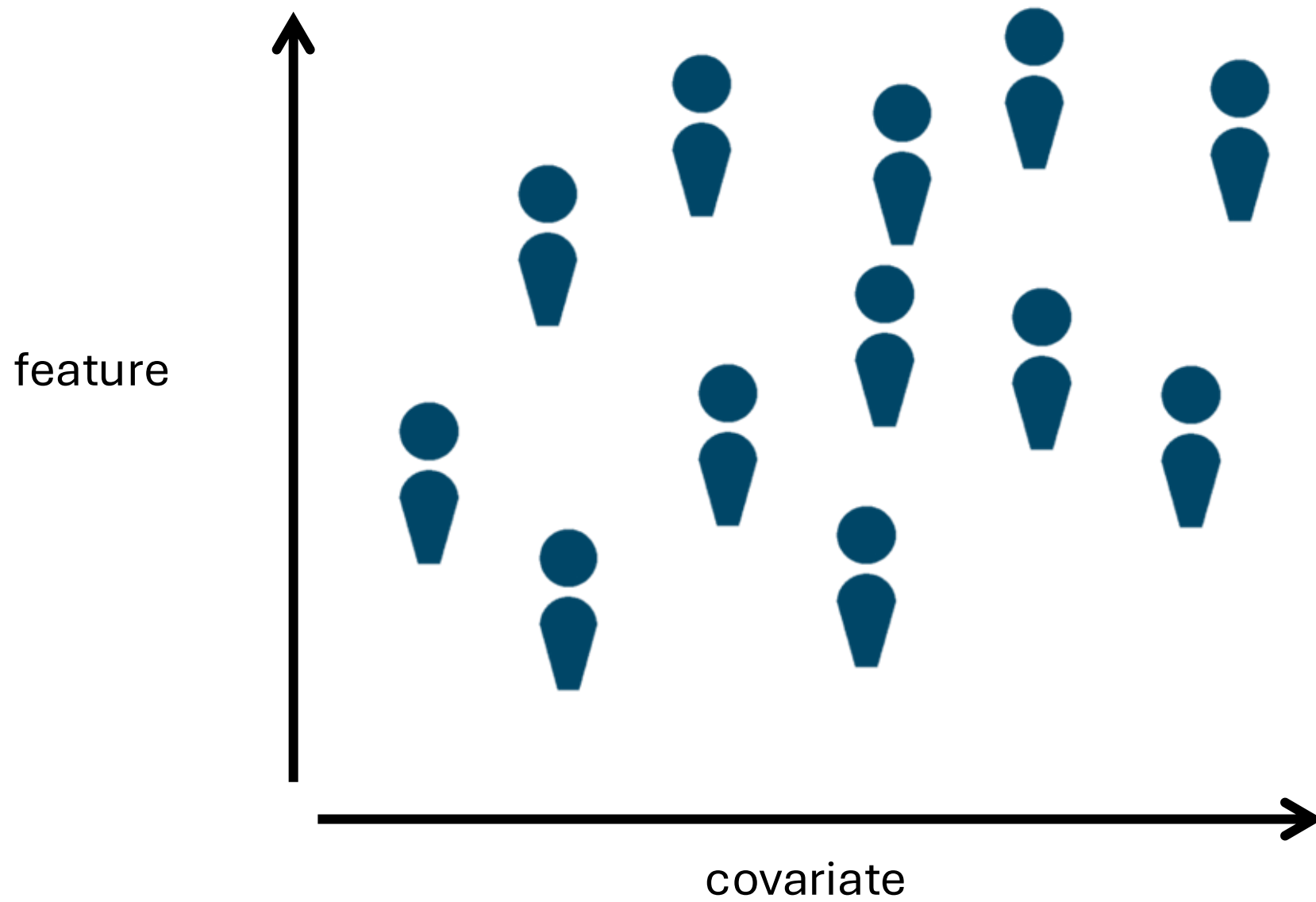
- Children know more with age
- Instead of absolute test scores, we compare them with respect to other children of their age

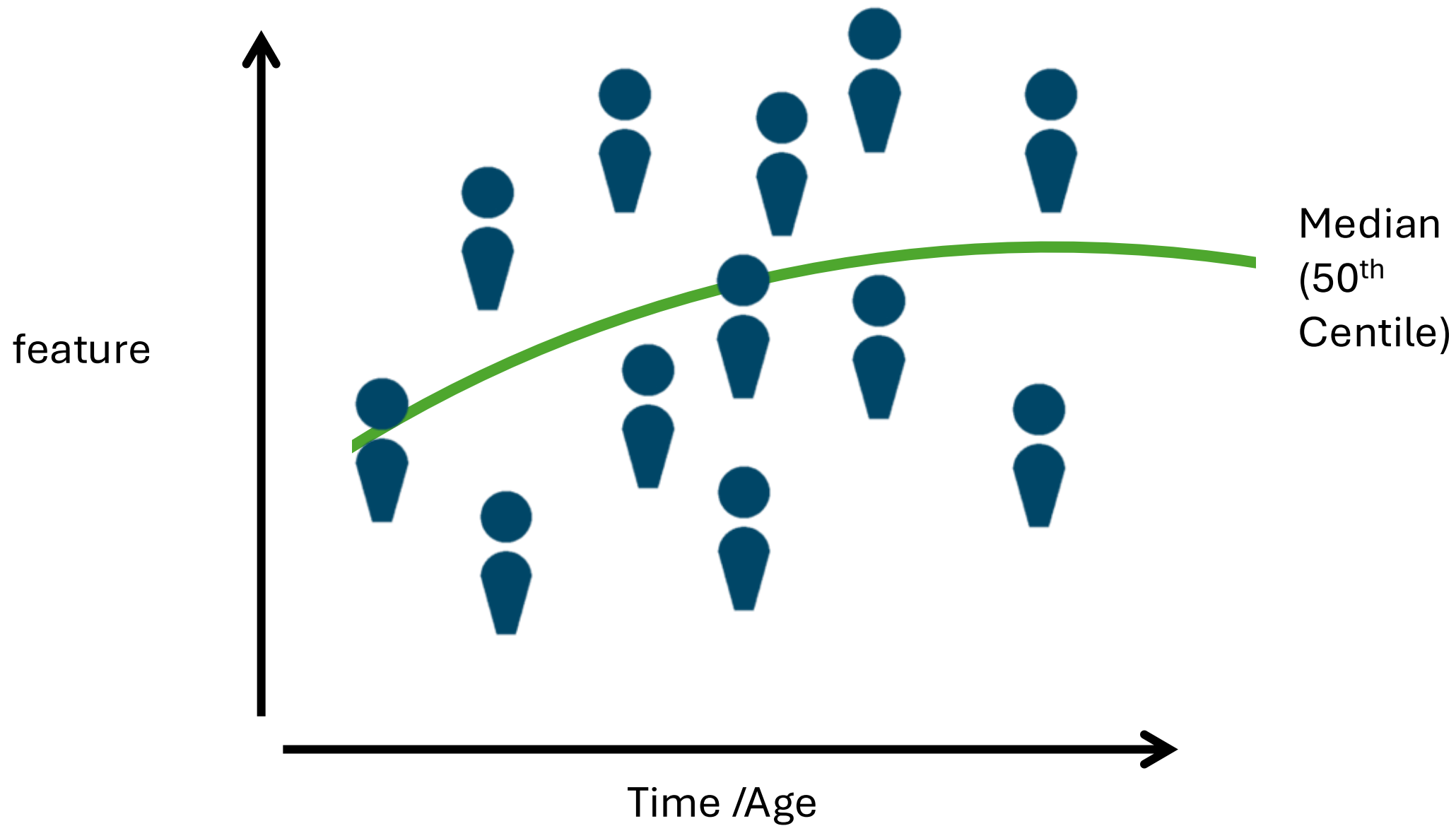


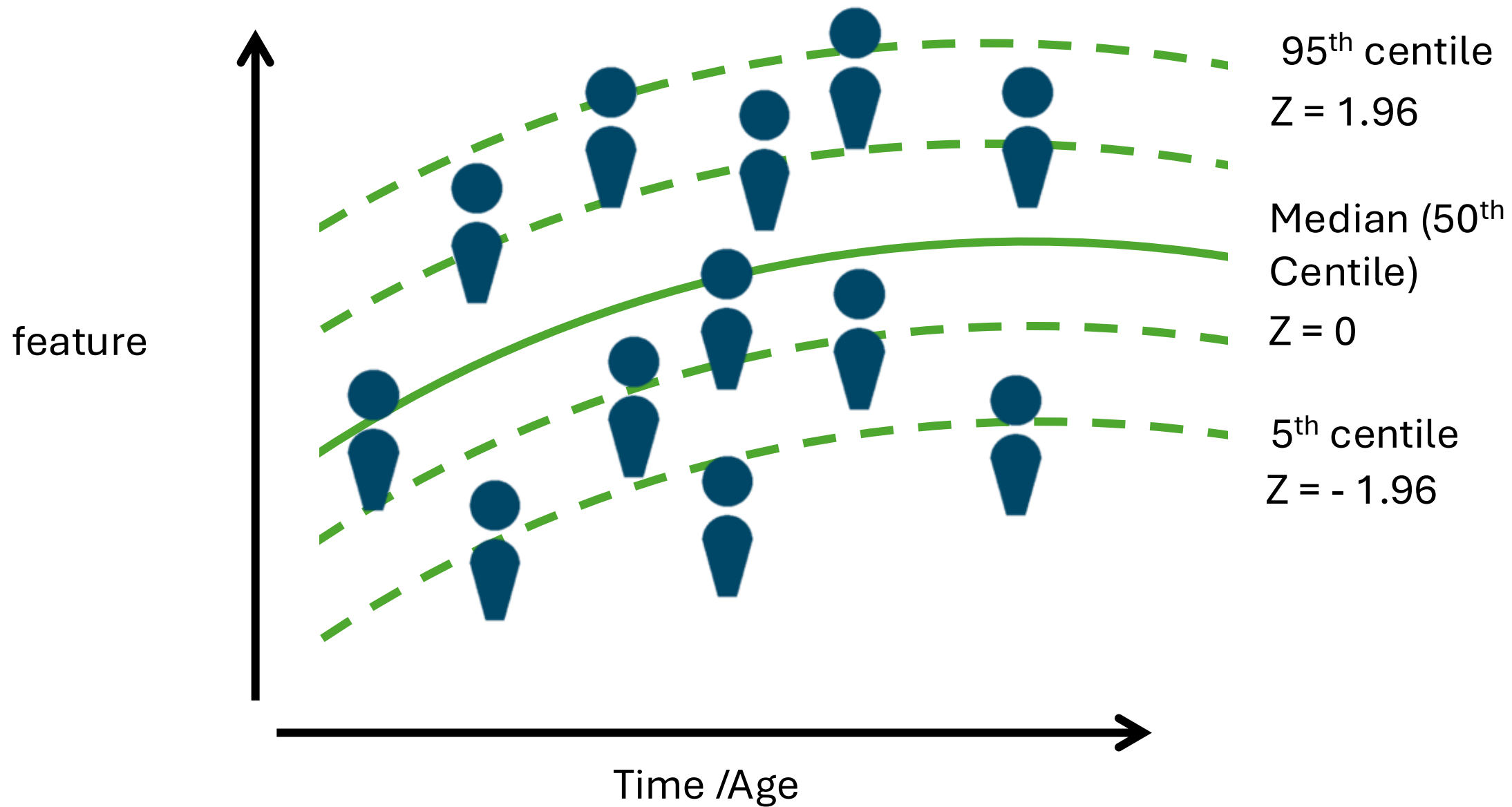
Score distribution chart for sample of 905 children tested on 1916 Stanford-Binet Test

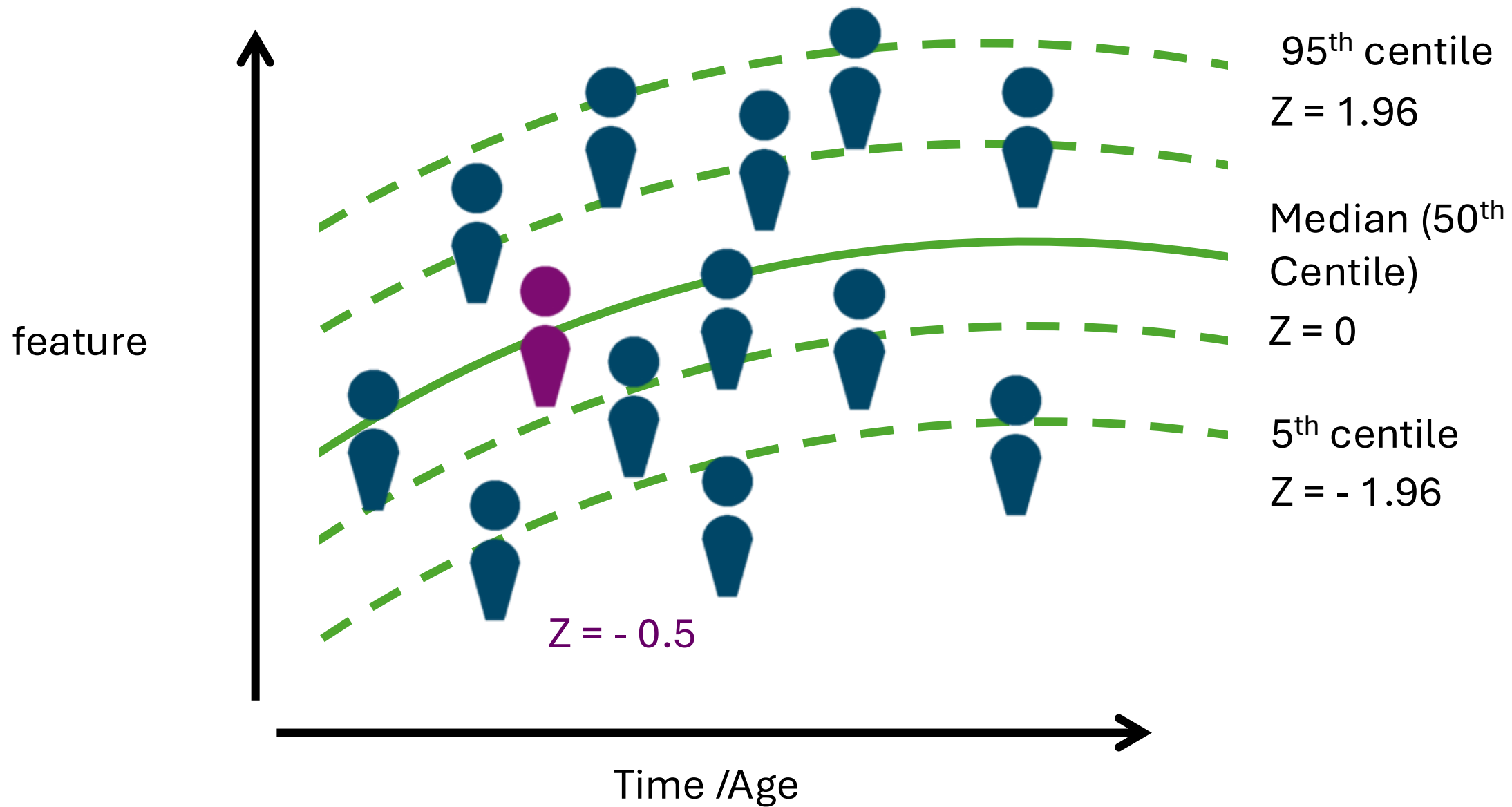
How are these charts created?











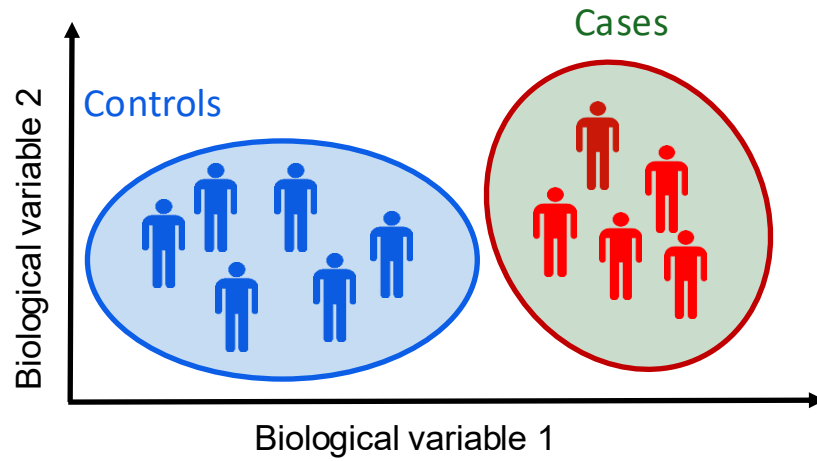
Why normative modelling in the life sciences?

- Why could this approach be useful?

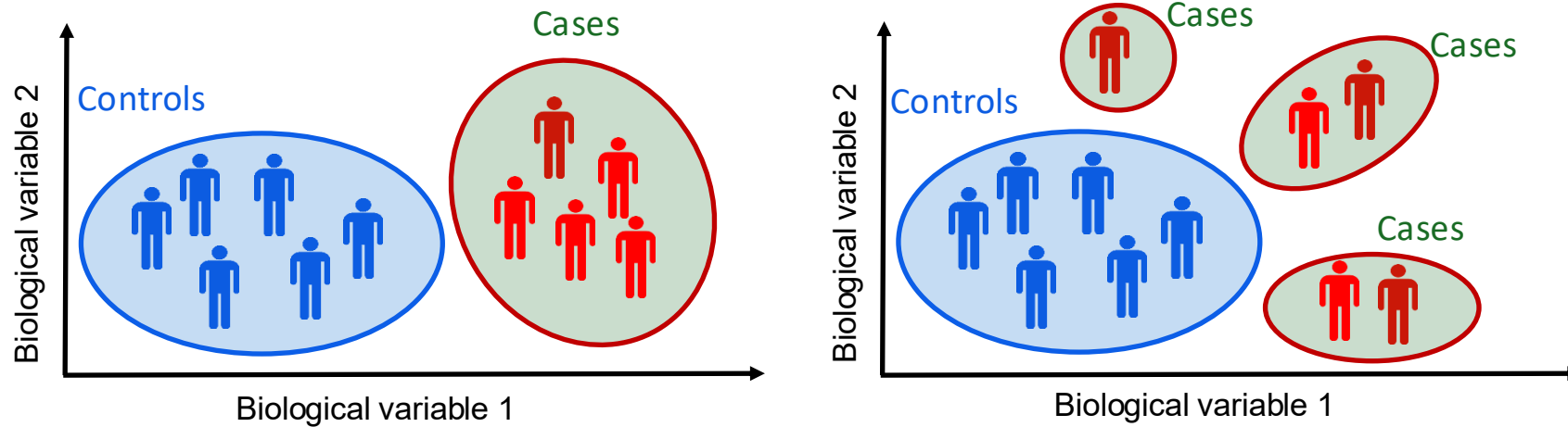
Why normative modelling?

- Biological systems are **inherently variable**: individuals, organisms, and populations differ in ways that group averages obscure (heterogeneity)

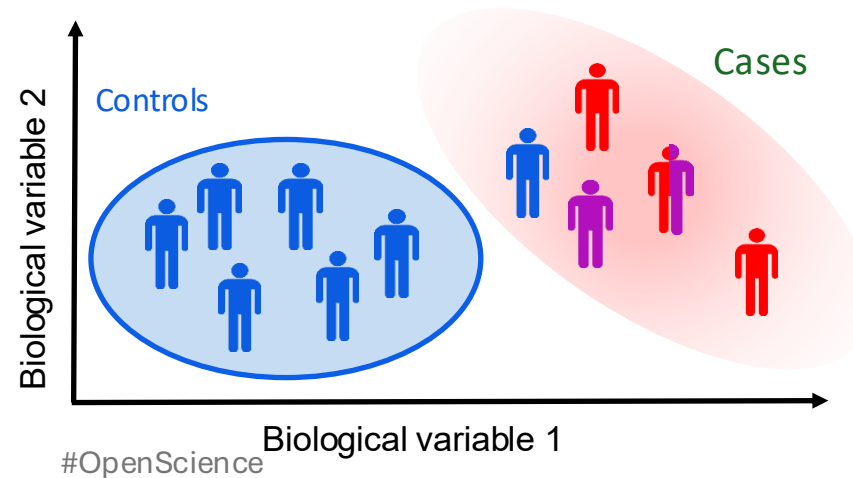
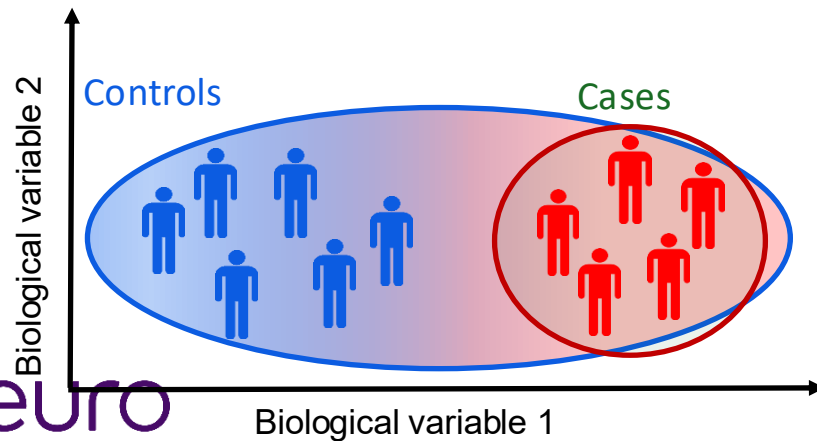
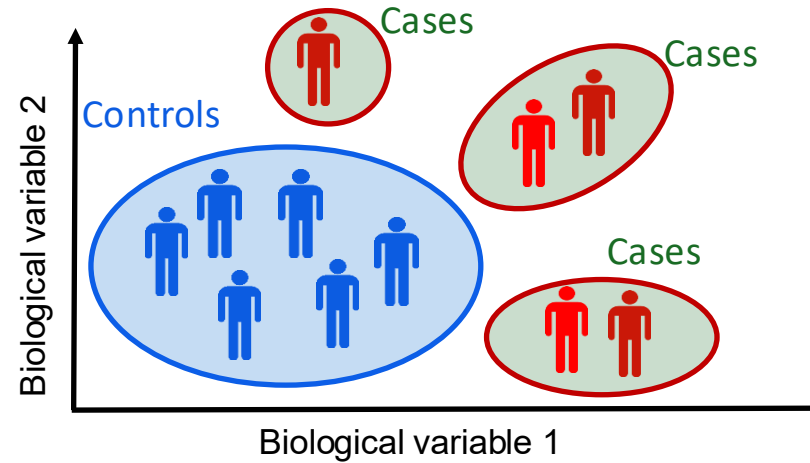
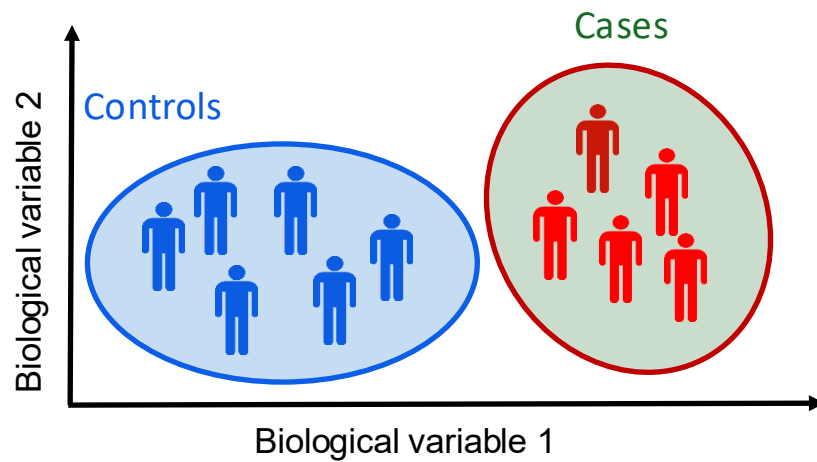
Beyond case control comparisons



Beyond case control comparisons



Beyond case control comparisons

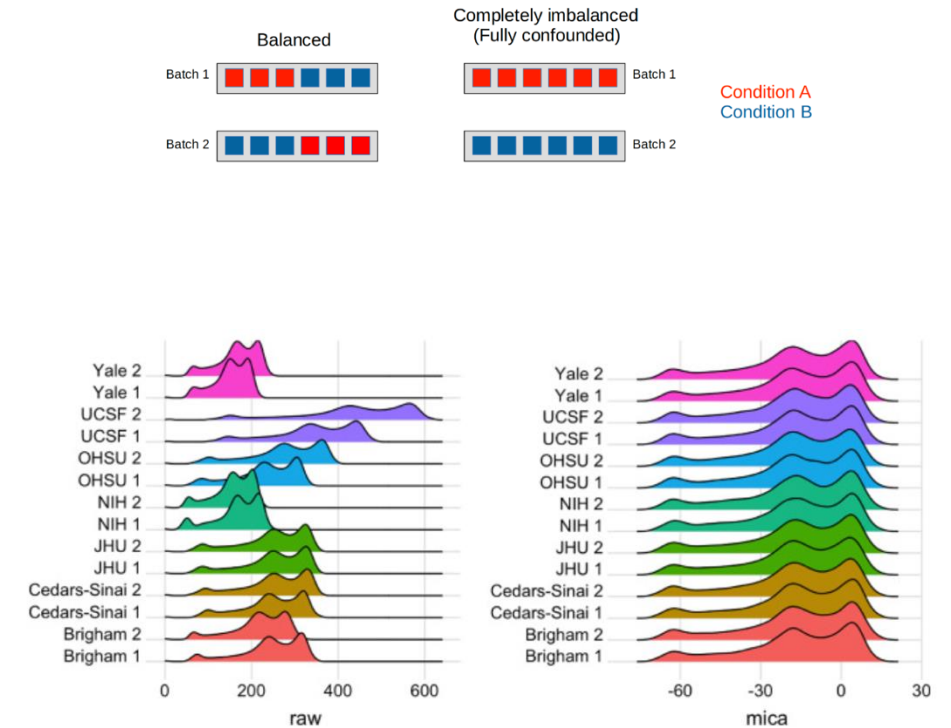


Why normative modelling?

- Biological systems are inherently variable: individuals, organisms, and populations differ in ways that group averages obscure (heterogeneity)
- Multi-site/batch and multi-instrument data introduces technical variation that confounds biological signal (Batch effects)

Batch effects

- non-biological, technical variations in data caused by samples being processed in different batches, such as
 - different days, labs, or technicians
 - Instruments, scanners, analysis pipelines
- introduce noise that can obscure biological signals, causing false positives or non-reproducible results, especially in genomics and single-cell studies

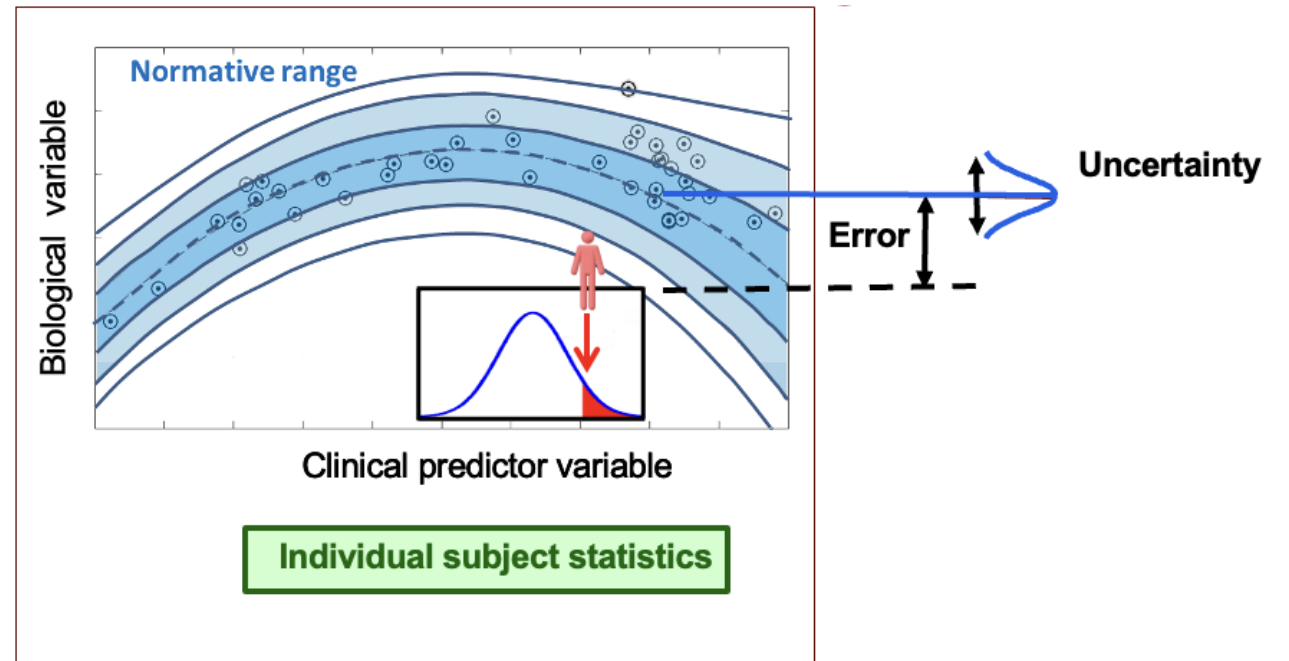


Why normative modelling?

- Biological systems are inherently variable: individuals, organisms, and populations differ in ways that group averages obscure (heterogeneity)
- Multi-site/batch and multi-instrument data introduces technical variation that confounds biological signal (Batch effects)
- Normative modelling separates genuine biological deviation from nuisance variation at the individual level

Two types of uncertainty

- Epistemic uncertainty
 - Modelling error
- Aleatoric uncertainty
 - Variability in the data

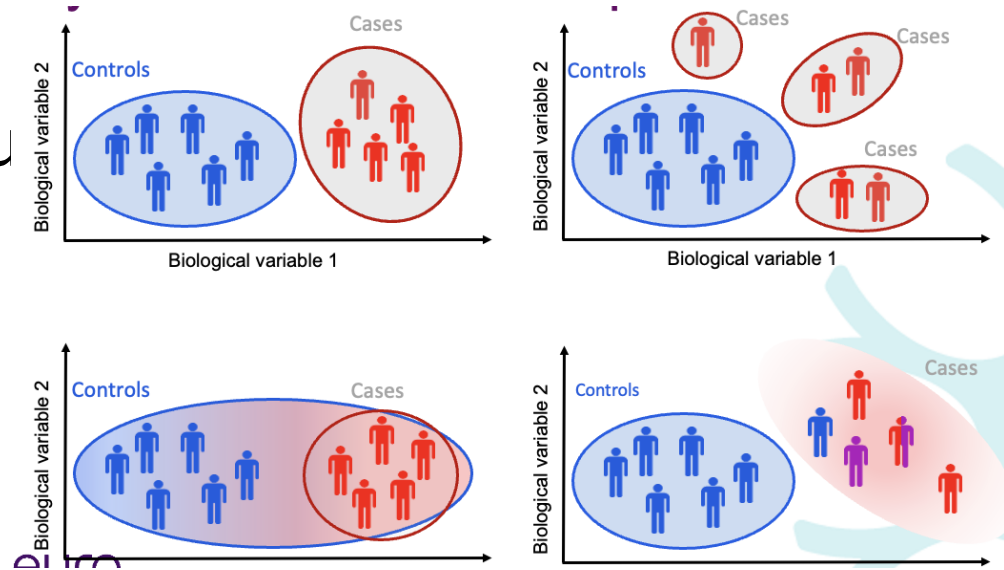


Why normative modelling?

Many classical statistical tests assume

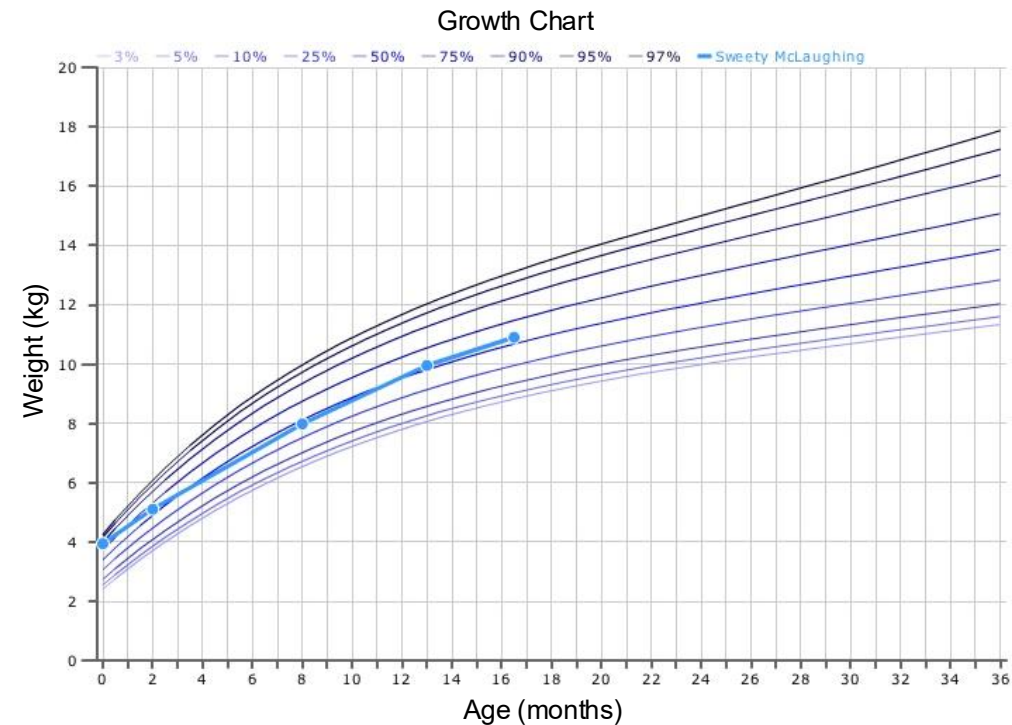
- Homogeneity of variances between groups
- Normal distributions of variables within each group
- No significant outliers in each group
- Independence of observations

Biological data routinely violates these: non-Gaussianity, heteroskedasticity, multi-site dependencies



Alternative: Normative Modelling

- 1. Estimate centiles of variation in a phenotype as a function of age, sex, and site
- 2. Assign each individual a Z-score relative to the control/reference distribution
- 3: New data point: Deviations from the norm capture individual-level heterogeneity rather than group averages



CASE

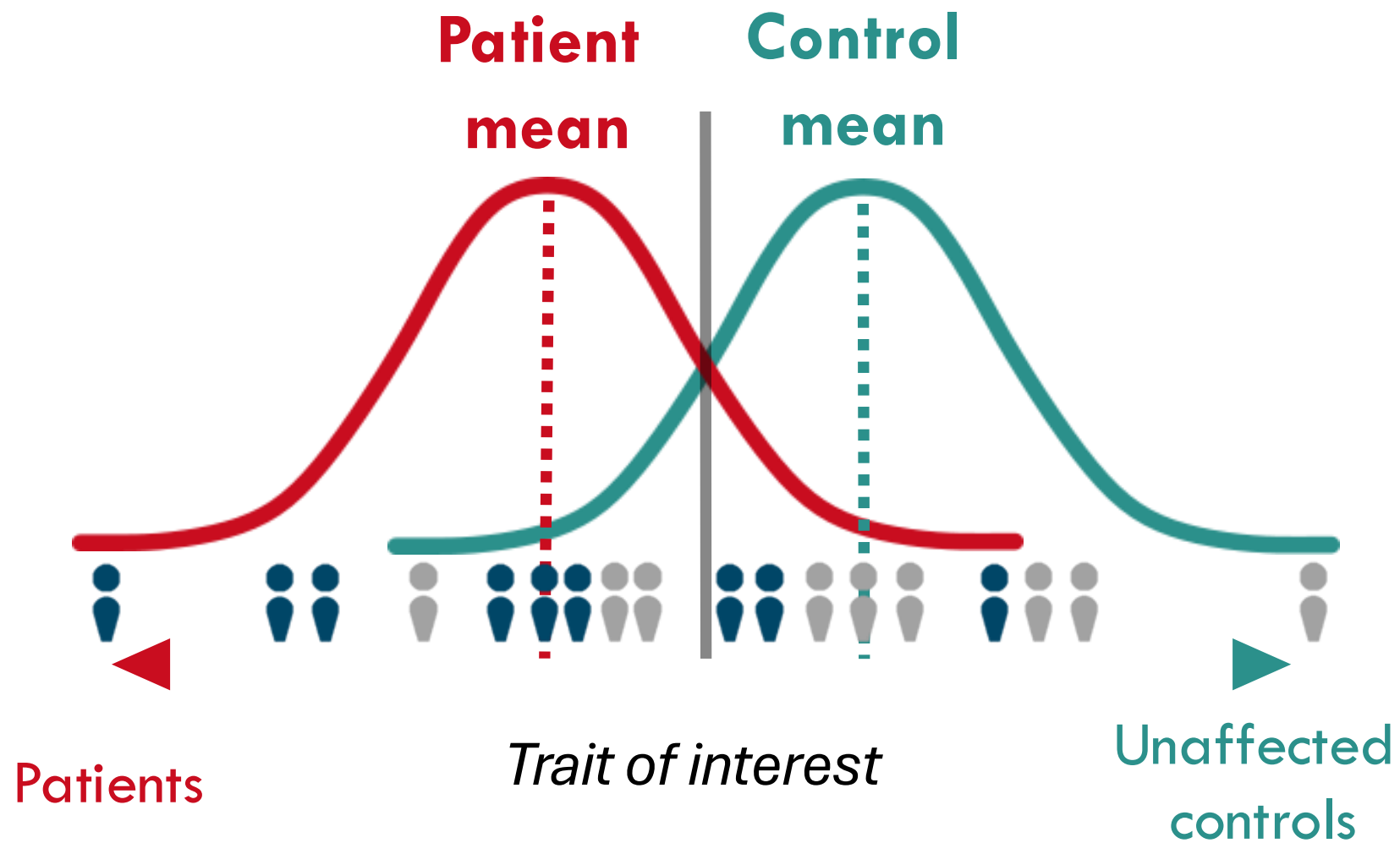


Patients

CONTROL



**Unaffected
controls**

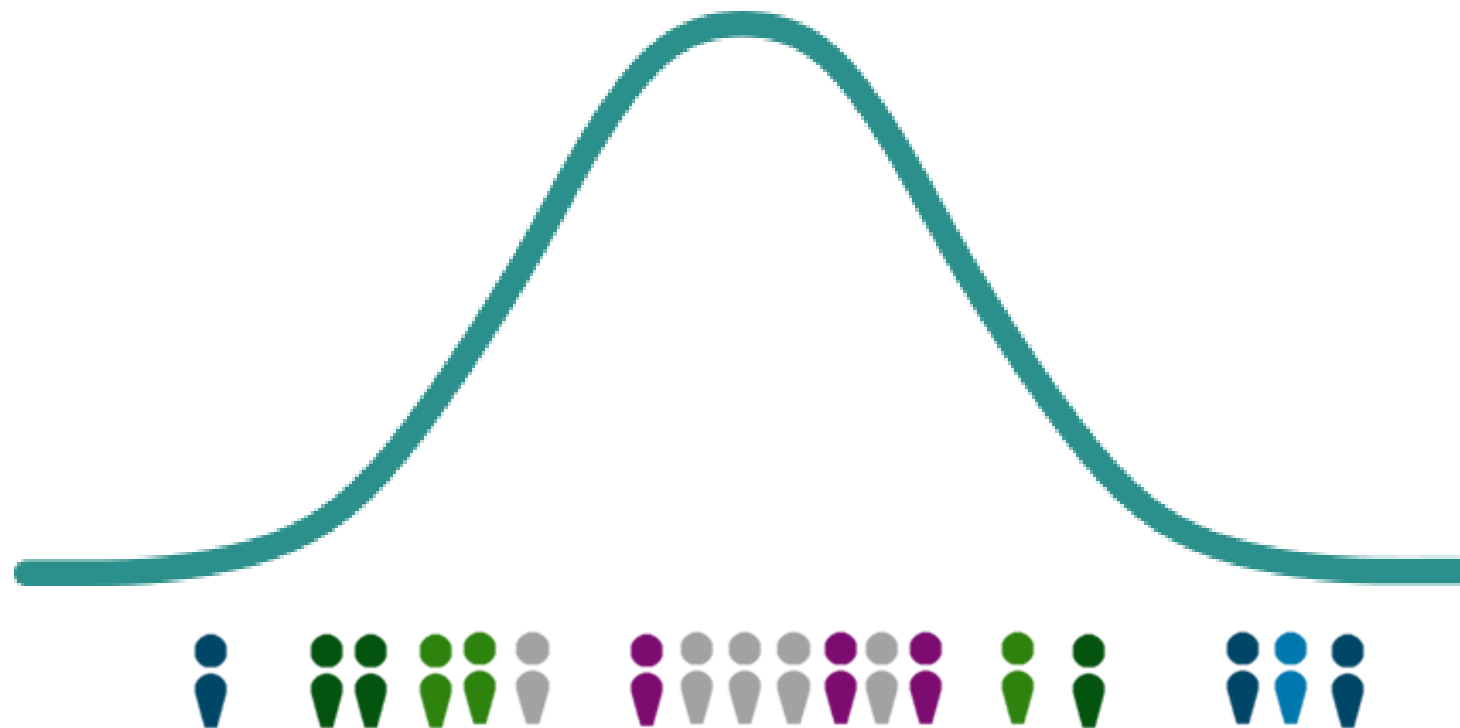


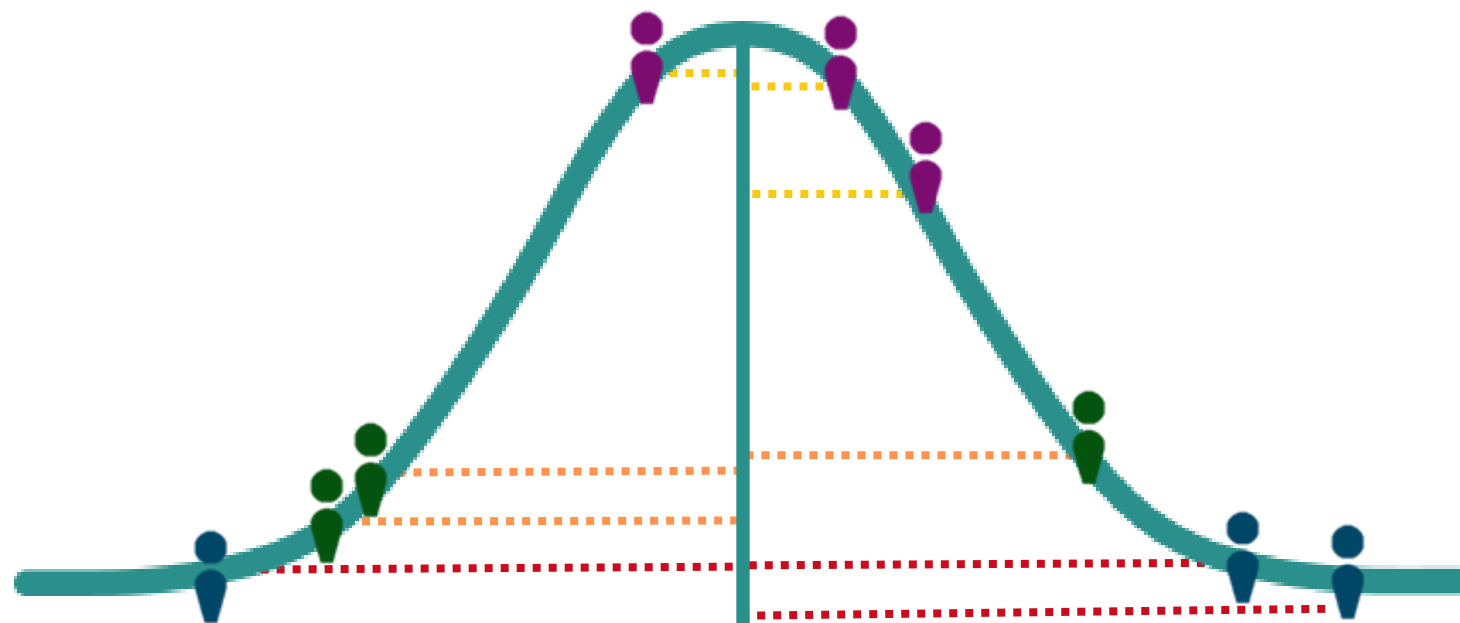


Patients



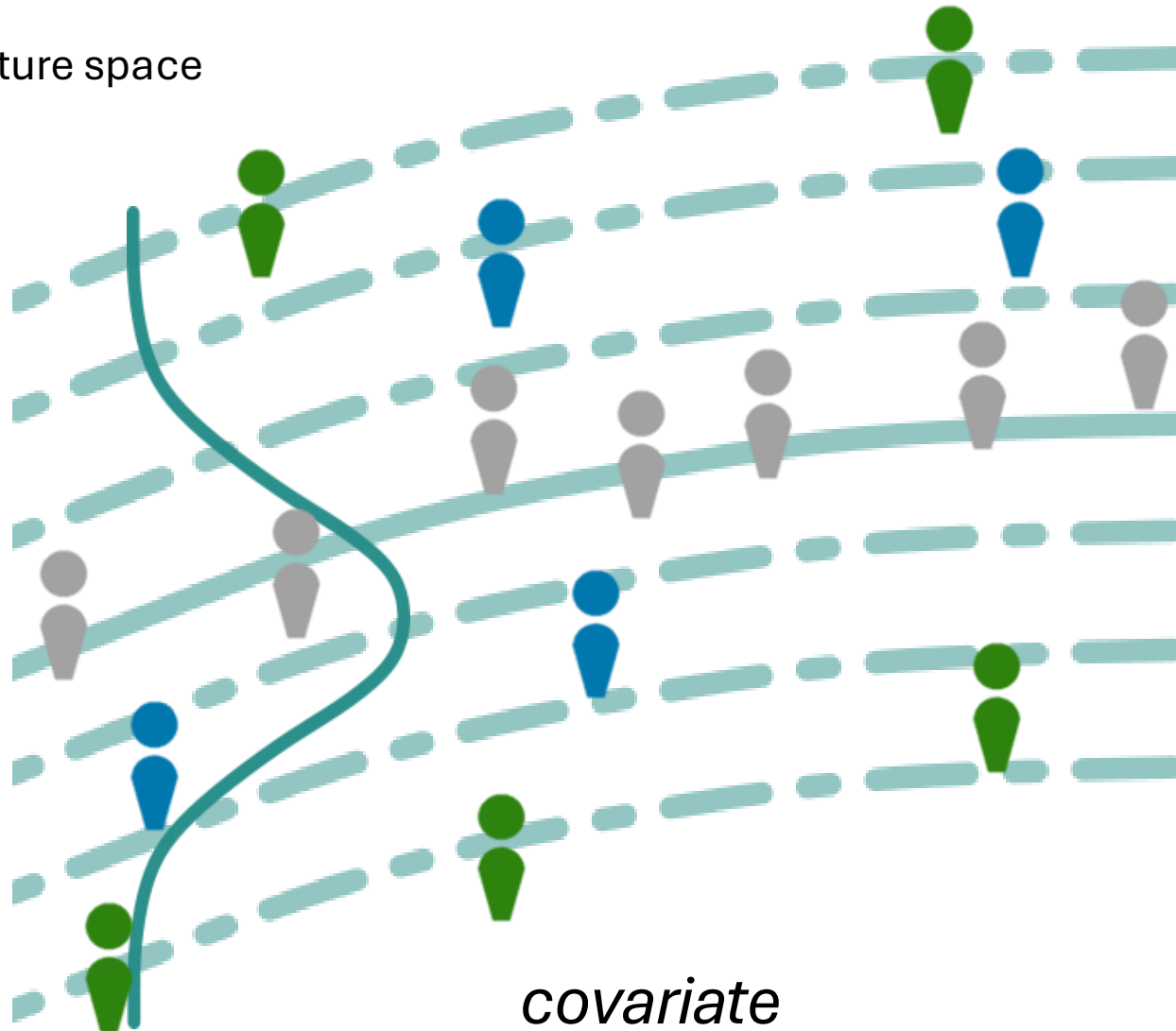
Unaffected
controls





Mapping into feature space

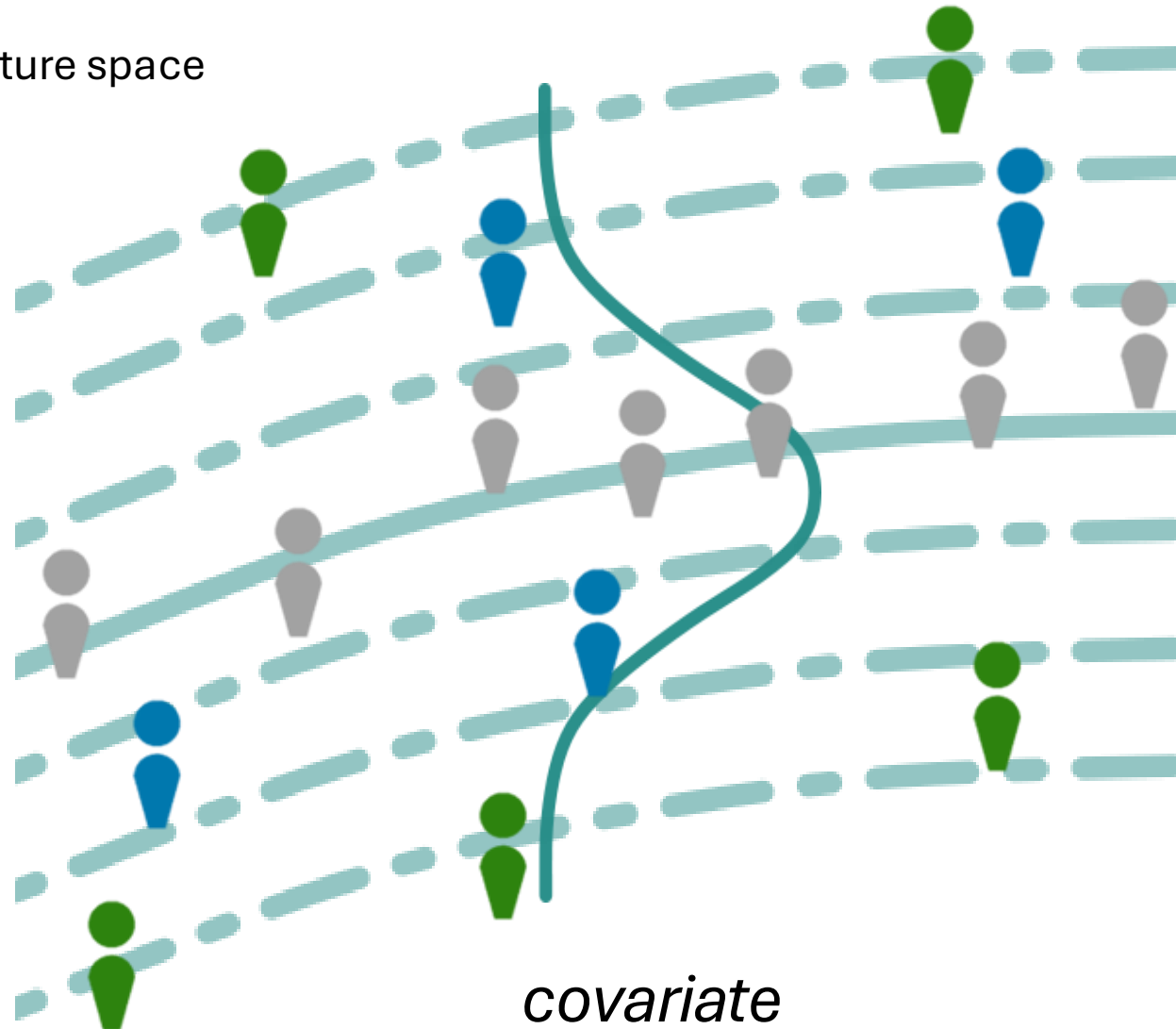
Feature of interest



covariate

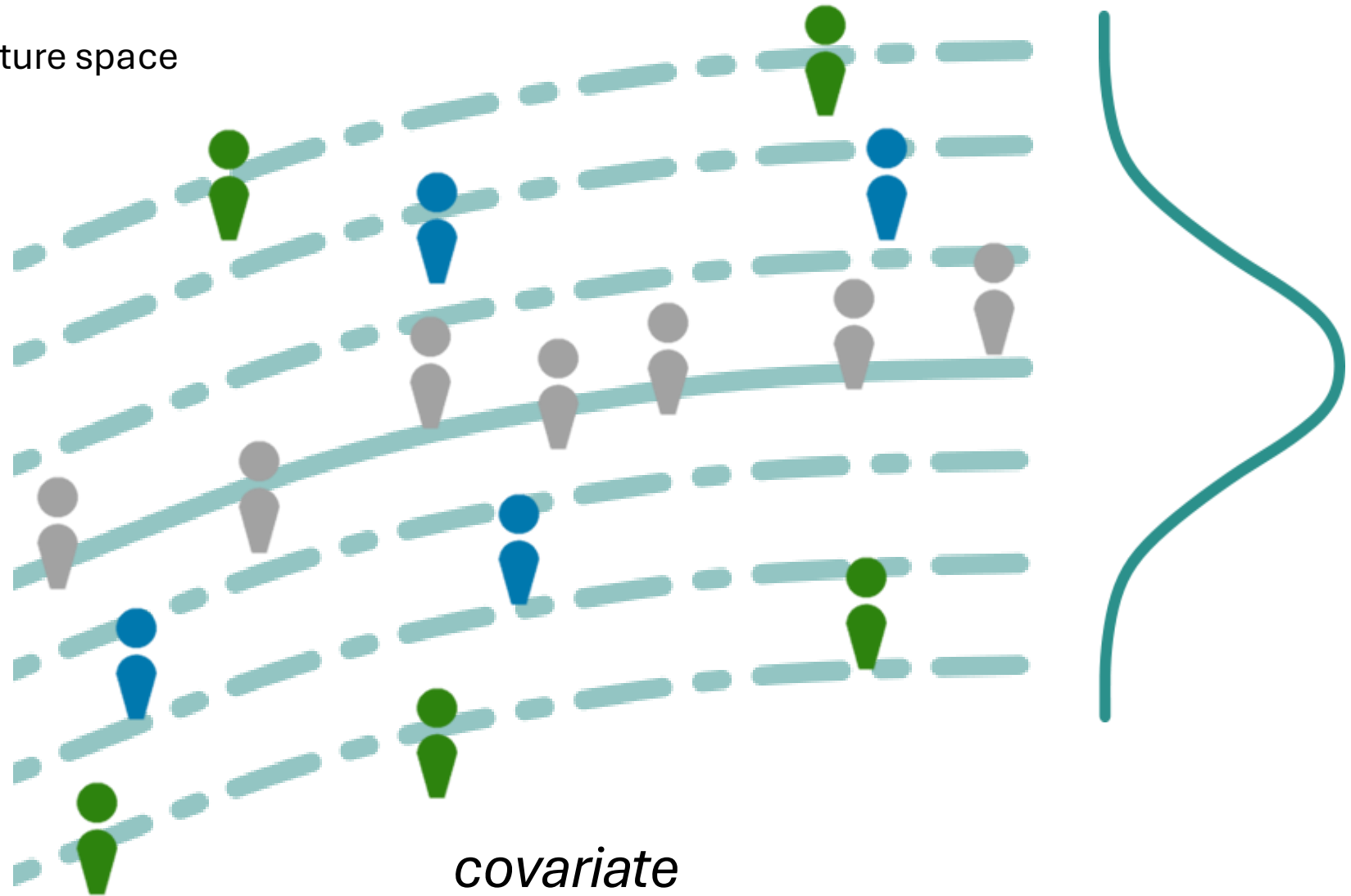
Mapping into feature space

Feature of interest

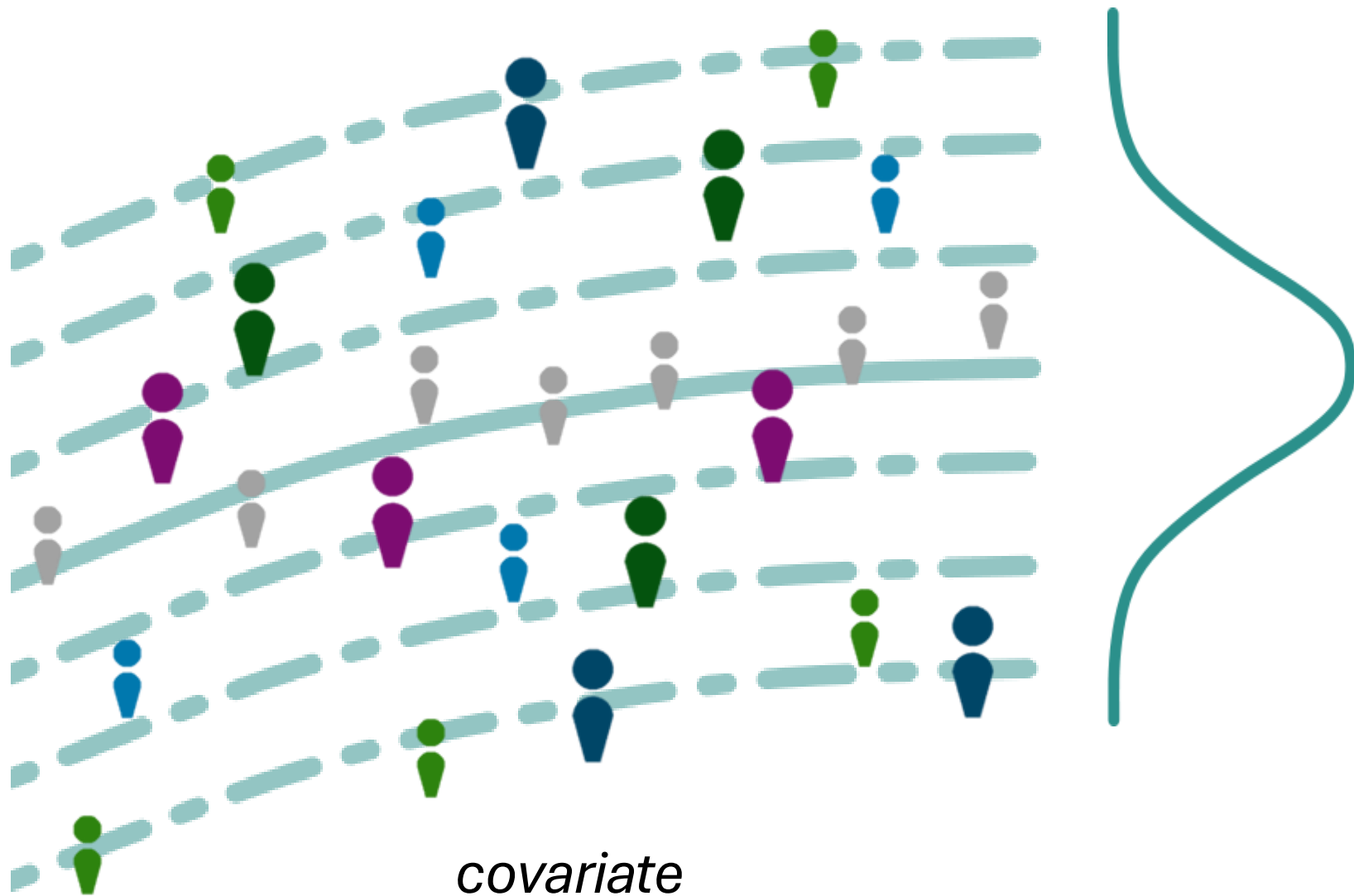


Mapping into feature space

Feature of interest

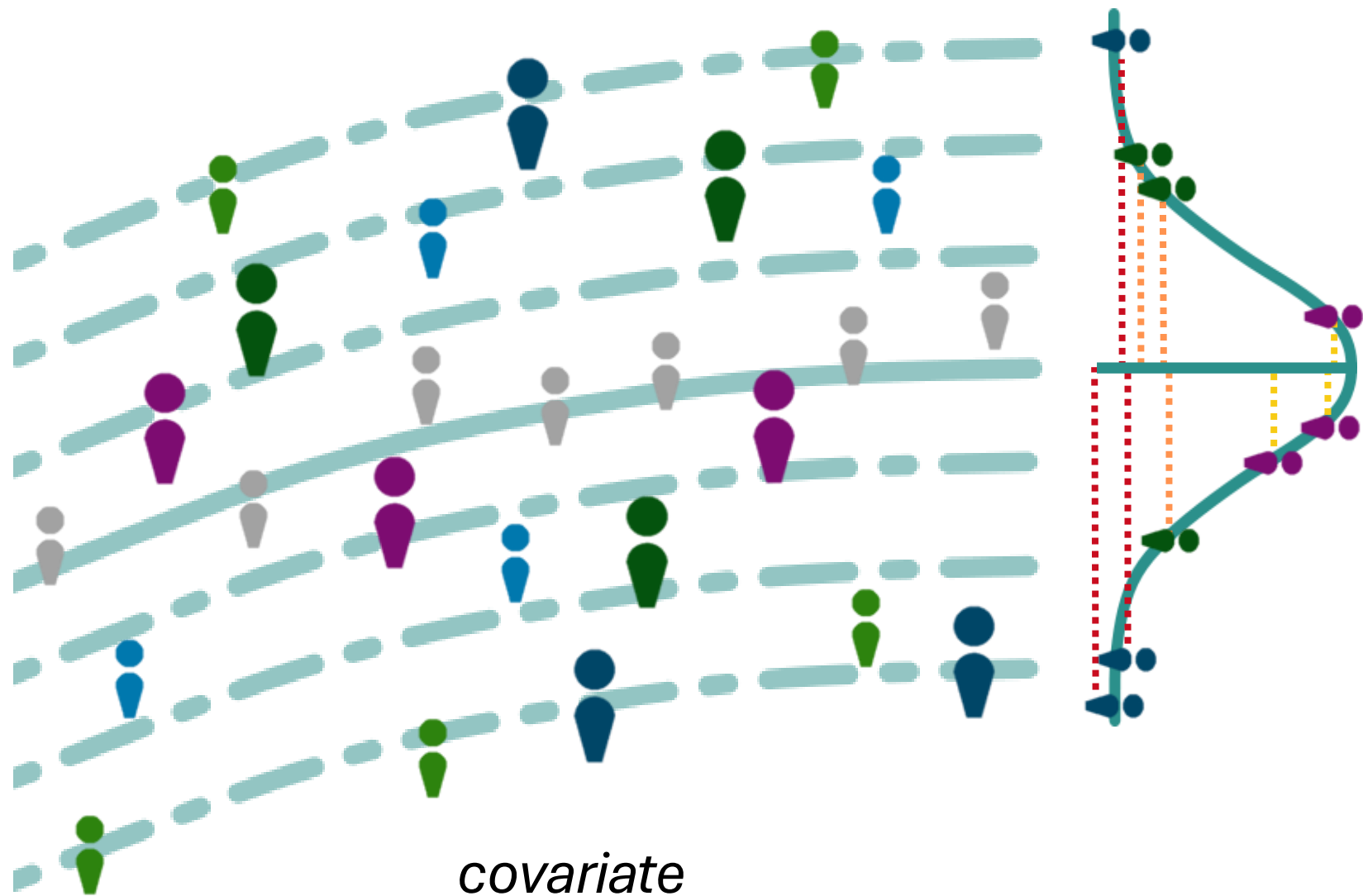


Feature of interest

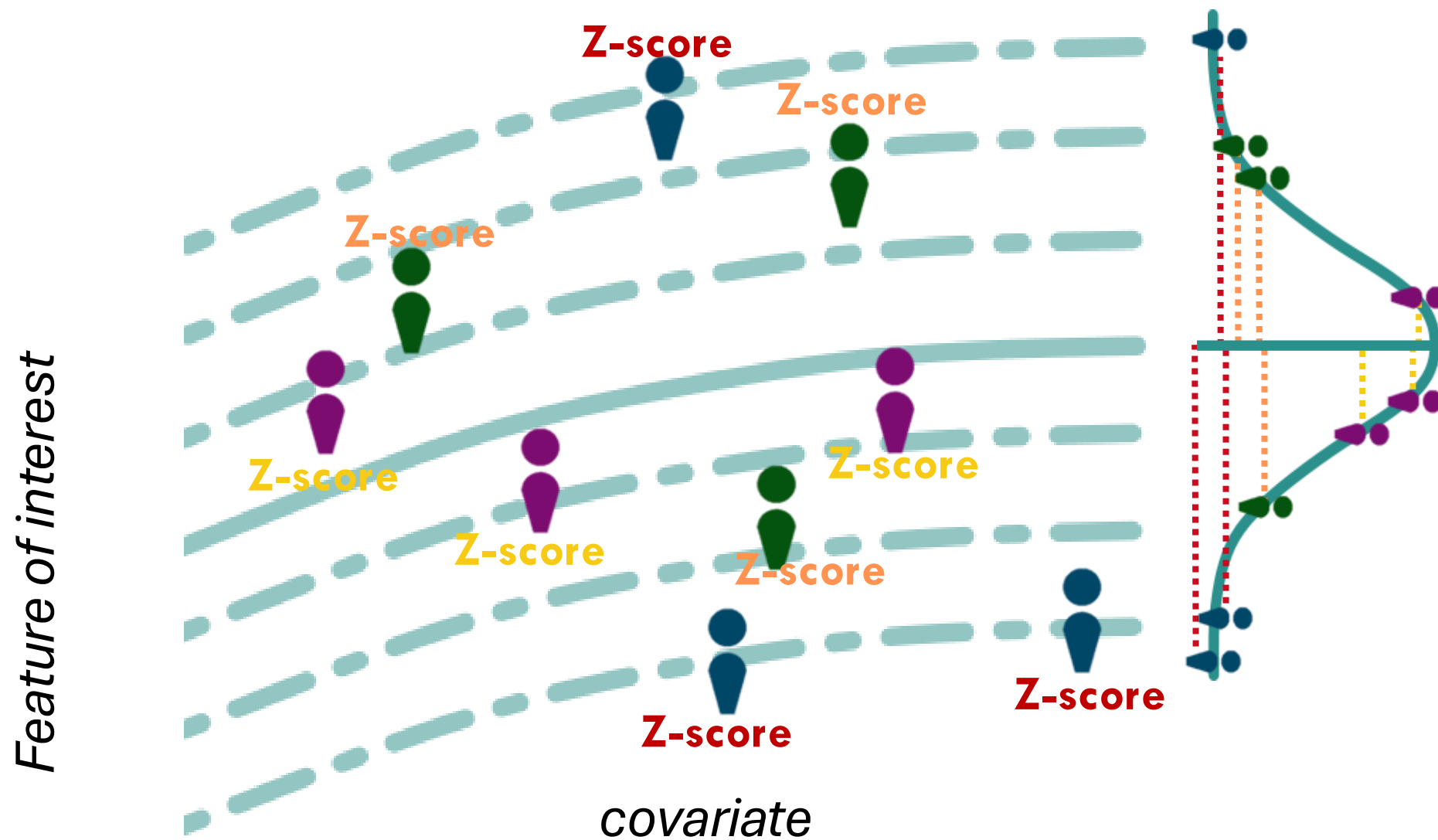


covariate

Feature of interest



covariate





Previous Applications

Normative modelling has been applied to:

- Neuroimaging: cortical thickness, connectivity across the lifespan
- Genomics: gene expression deviations in disease (Fraze et al., 2024)
- Ecology: climate, pollution, urbanicity, regional socioeconomic conditions (ENVIRONMENTAL consortium)
- Developmental biology: growth trajectories in model organisms (Bayer et al., 2026)
- Data harmonisation: Site and batch effect correction (Bayer et al., 2022)

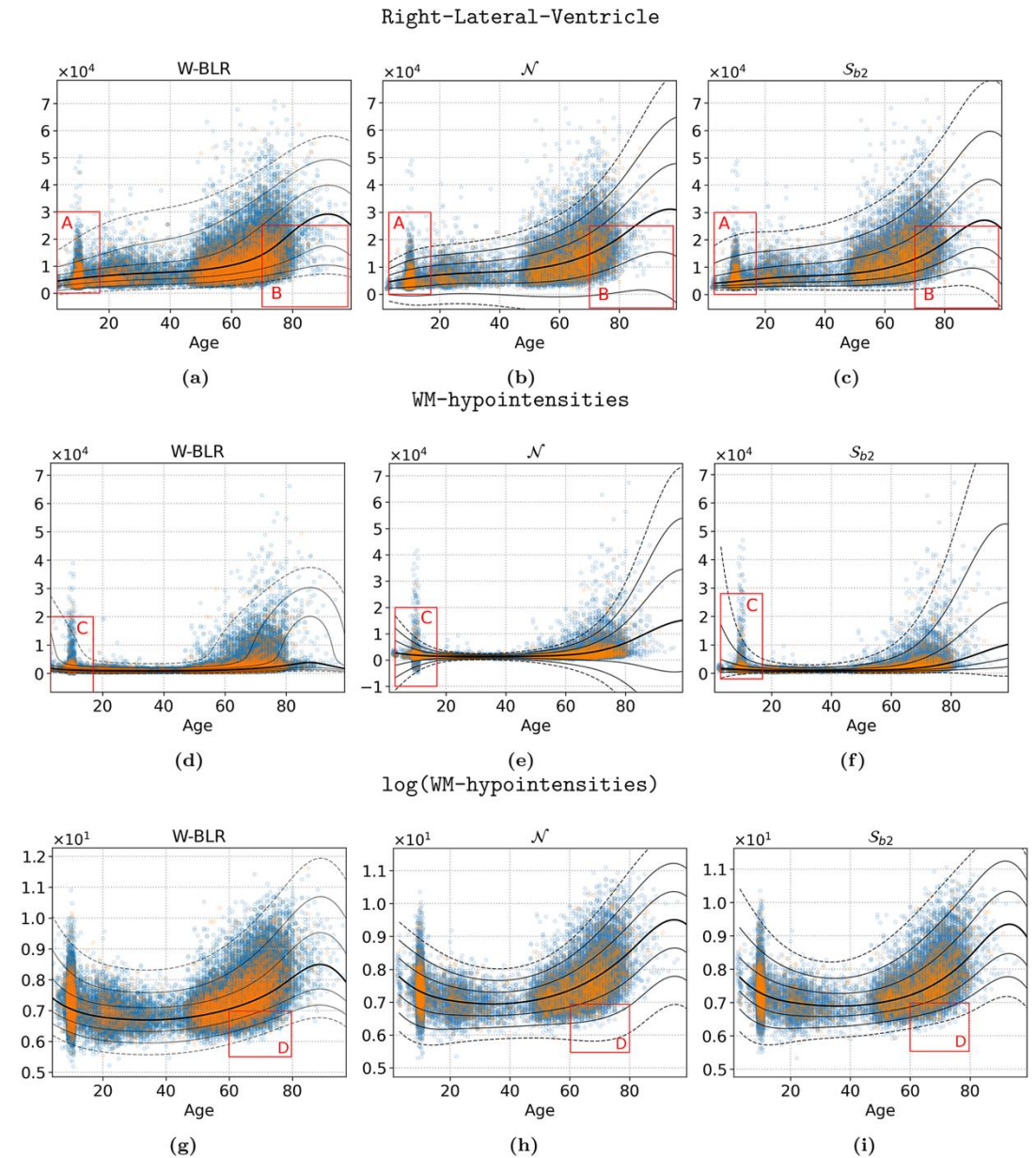
Structural Neuroimaging

- Freesurfer
- Brain imaging software tool
- Allows the extraction of features of the brain
- Example:
 - Segmentation of brain into regions according to atlas
 - Extraction of the average thickness of the cortical sheet of that region
 - Desikan Atlas: 34 bilateral regions, one bilateral whole brain average



Structural Neuroimaging

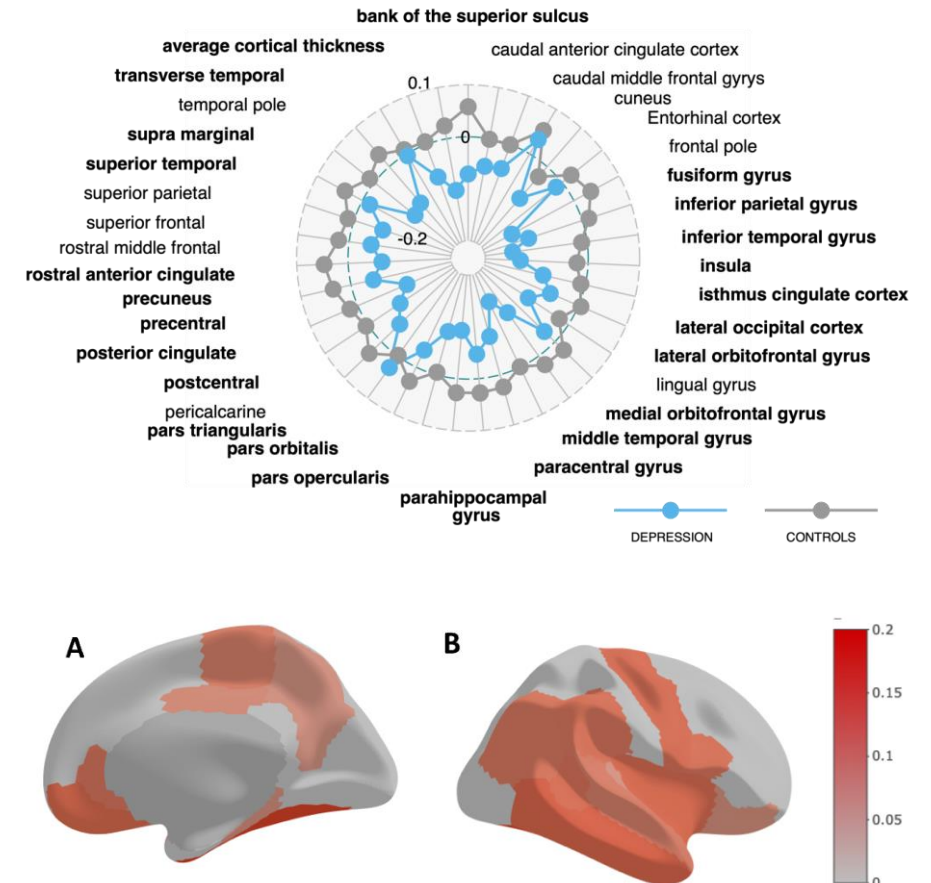
- Normative models derived from 40000 brain scans
- the Right Lateral Ventricle,
- WM hypointensities
- log WM hypointensities



Structural Neuroimaging

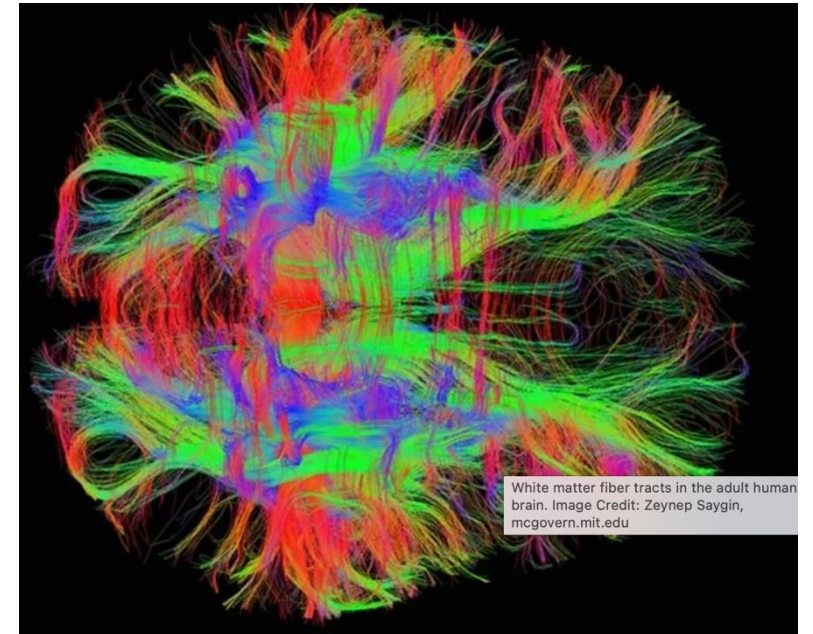
- Create one normative model per region
- How do groups of interest differ in their amount of individuals with extreme deviations?
- Count the percentage of extreme deviations

Average z-score difference

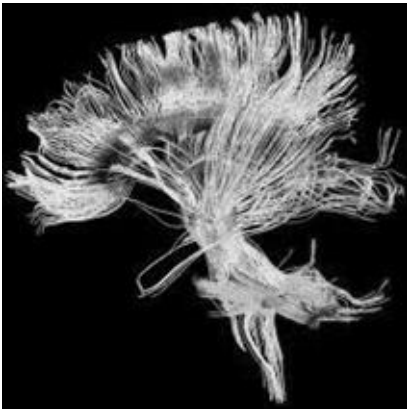


Diffusion Tensor Imaging

- maps the brain's white matter tracts by measuring the direction of water molecule diffusion
- Common Measures:
 - Fractional Anisotropy:
 - Measures the direction of diffusion
 - Mean Diffusivity:
 - Measures the mean diffusion



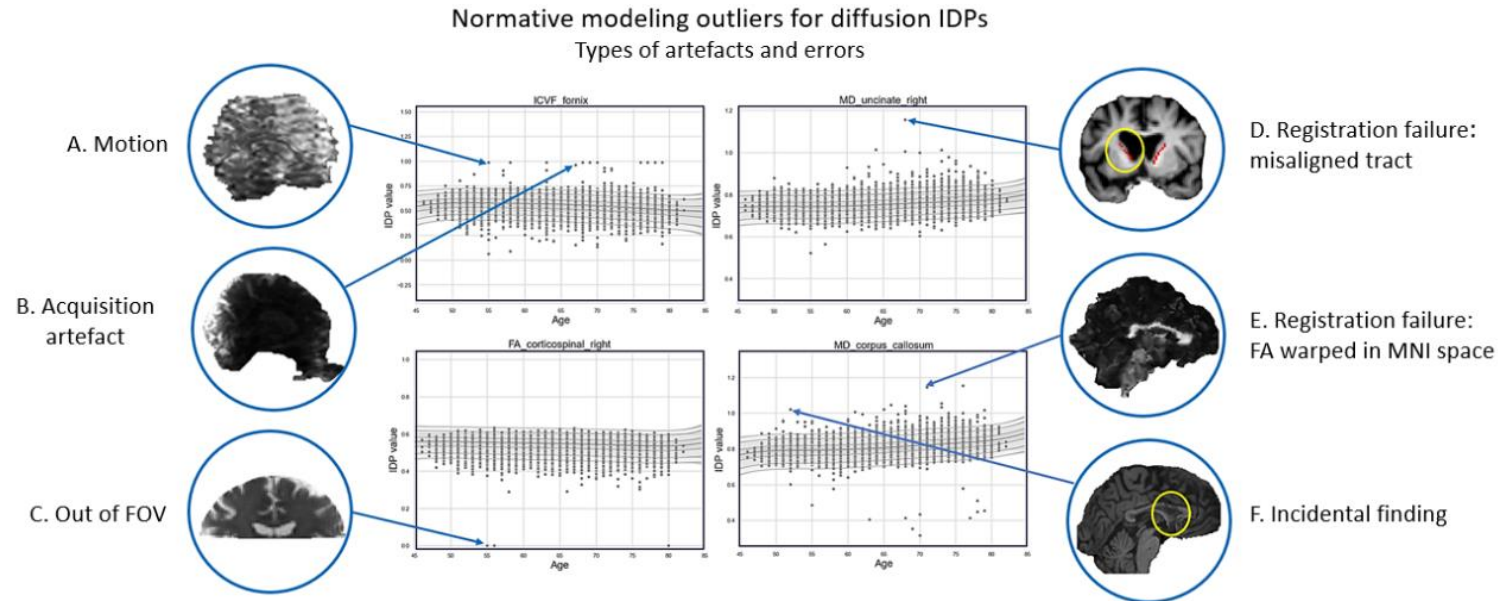
Diffusion Tensor Imaging



DTI

Ramona Cirstian

ramona.cirstian@donders.ru.nl



Psychometrics

PRECOGNITION

Learning Latent Cognitive Profiles to Predict Psychosis

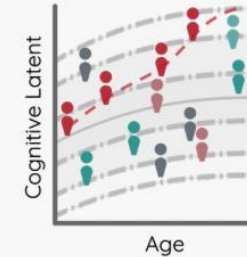


Psychometrics



Focus groups

This project is dedicated to including individuals with **lived experiences of psychosis**, a commitment reflected in our team which comprises experts with firsthand knowledge of psychosis. The cornerstone of our research is in organizing focus groups to understand the lives of people living with psychosis and identify how cognition impacts their daily functioning.



Lifespan Cognitive Models

Driven by the information from focus groups, we aim to **develop lifespan cognitive reference models** that capture various aspects of cognition, such as processing speed, working memory, and verbal learning, among others. These models will be based on existing data resources of tens of thousands of participants across Europe.

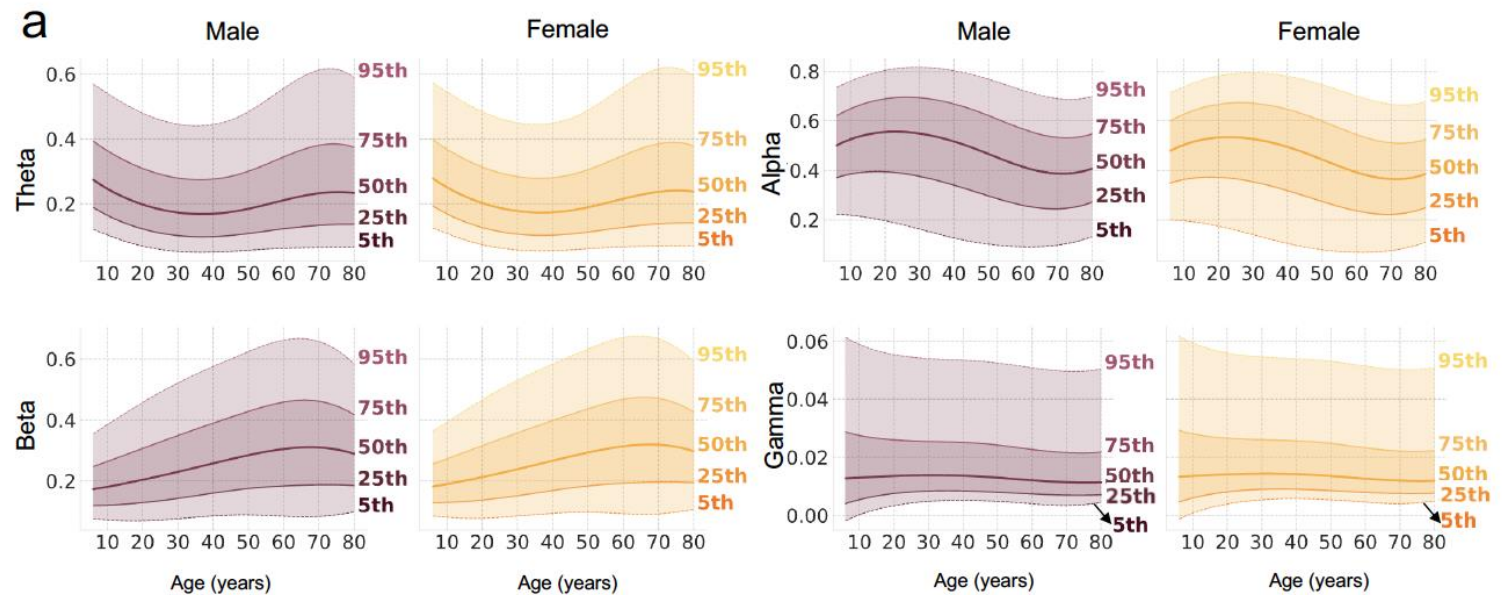


Individual Outcomes

Our overall aim is to **predict functional outcomes in individuals with psychosis** by using the cognitive lifespan models. These models will enable us to understand the progression of the disorder at an individual level and predict the early onset of psychosis. Using our framework, we hope to take the first steps towards individualized disease prediction and understanding.

Electro and Magnetencephalography

- Normative models of MEG frequency bands across age

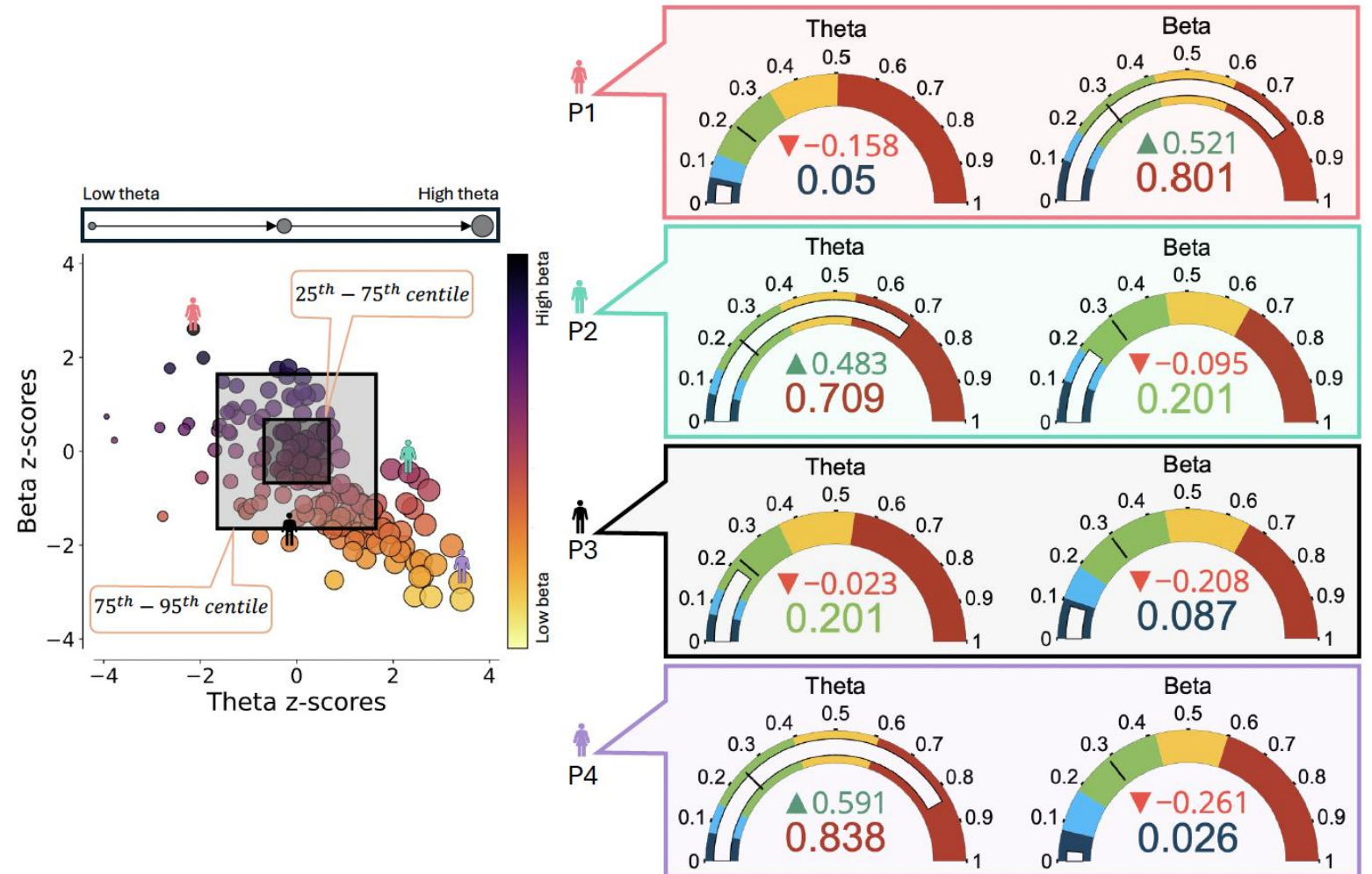


Zamanzadeh, M., Verduyn, Y., de Boer, A., Ros, T., Wolfers, T., Dinga, R., Šafář Postma, M., Marquand, A. F., van Wingerden, M., & Kia, S. M. (2026). Normative modeling of MEG brain oscillations across the human lifespan. *Communications Biology*, 1–16.

Electro and Magnetencephalography

c

- Theta and Beta frequency activation for individual patients, mapped on the population



Harmonisation

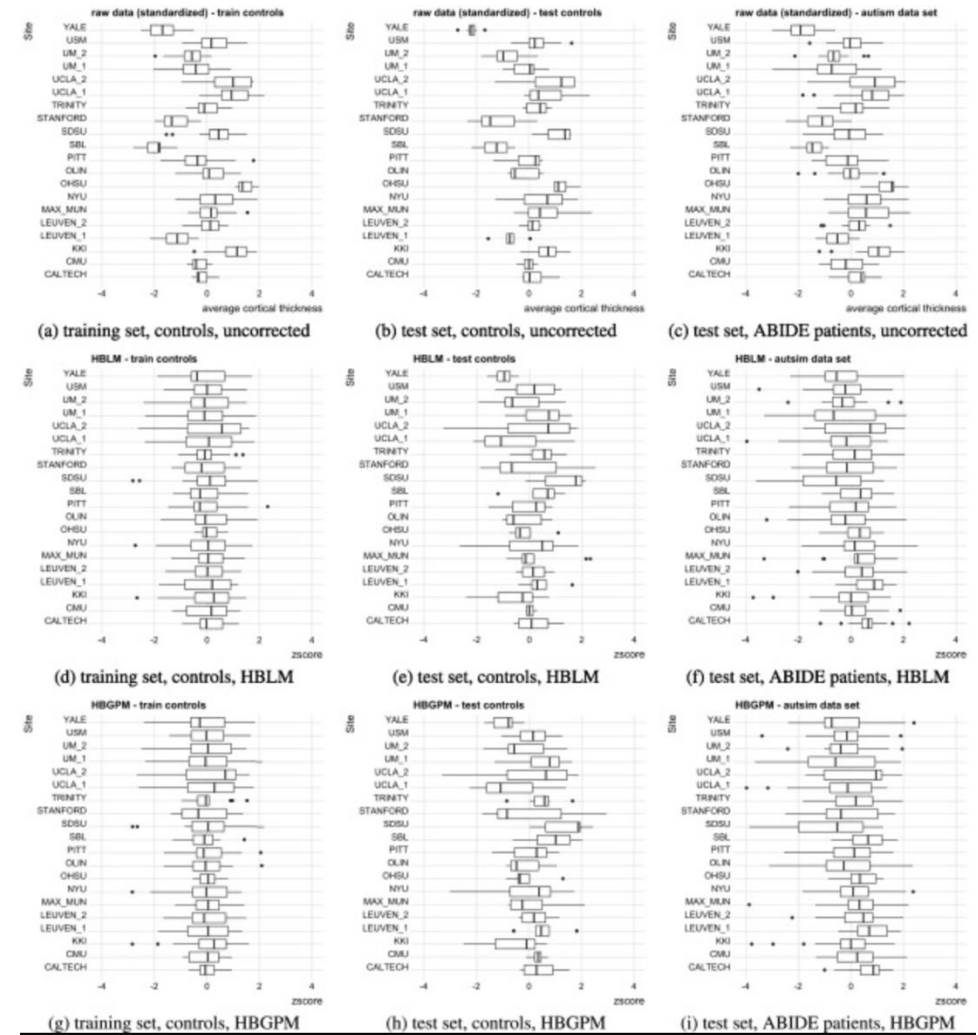


Harmonisation



Batch variable

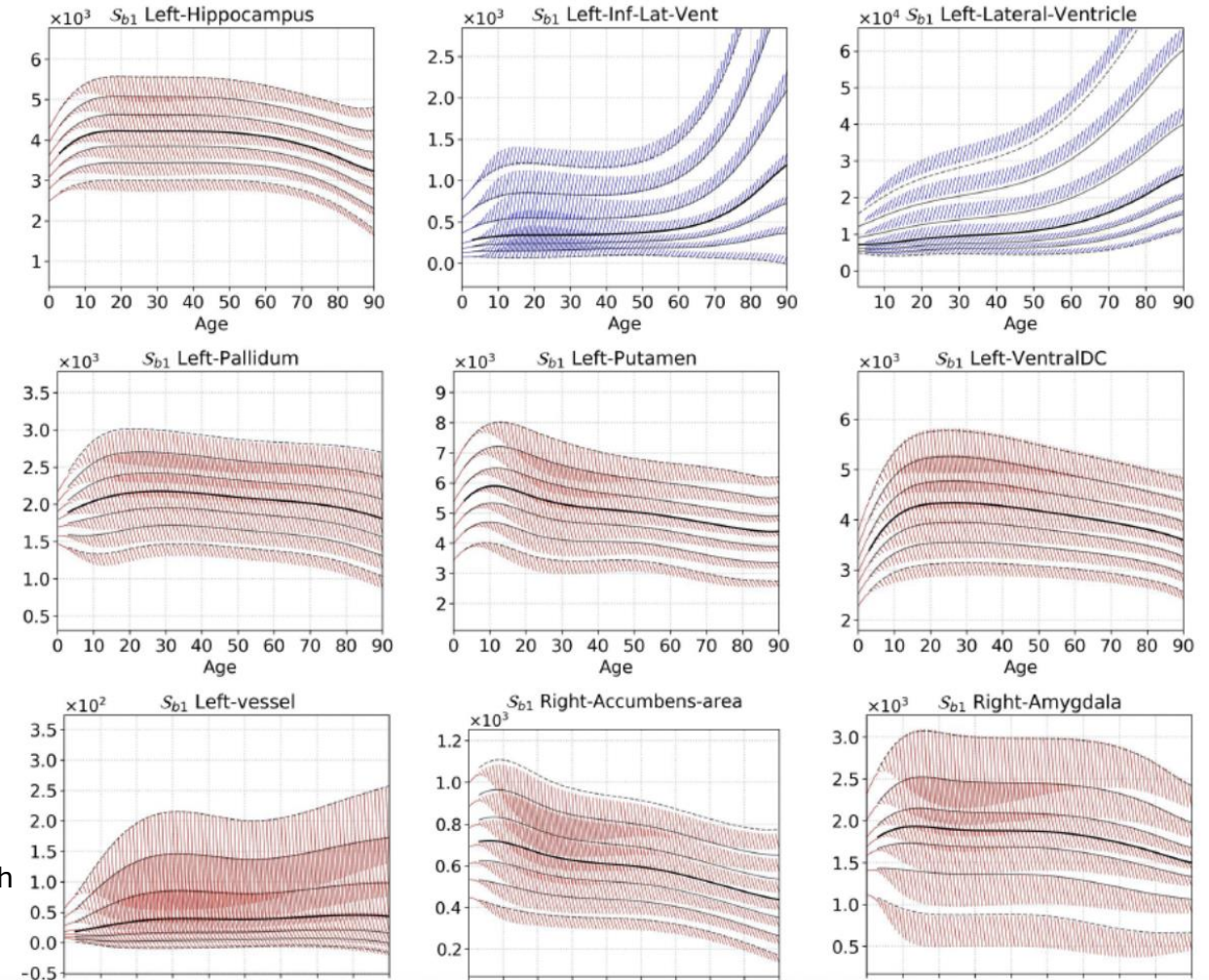
- Including Site as a batch effect into a normative model



Bayer, J. M. M., Dinga, R., Kia, S. M., Kottaram, A. R., Wolfers, T., Lv, J., Zalesky, A., Schmaal, L., & Marquand, A. (2022). Accommodating site variation in neuroimaging data using normative and hierarchical Bayesian models. *NeuroImage*, 264, 119699.

Longitudinal normative models

- Charting the speed of brain change



Bayer, J. M. M., de Boer, A. A. A., Rehak-Bucova, B., Fraza, C. J., Banaschewski, T., Barker, G. J., Bokde, A. L. W., Bruehl, R., Desrivieres, S., Flor, H., Garavan, H., Gowland, P., Grigis, A., Heinz, A., Lemaitre, H., Martinot, J.-L., Martinot, M.-L. P., Artigues, E., Nees, F., ... Marquand, A. F. (2026). Charting the velocity of brain growth and development. In *arXiv [q-bio.NC]*. arXiv. <https://doi.org/10.48550/arXiv.2601.07591>

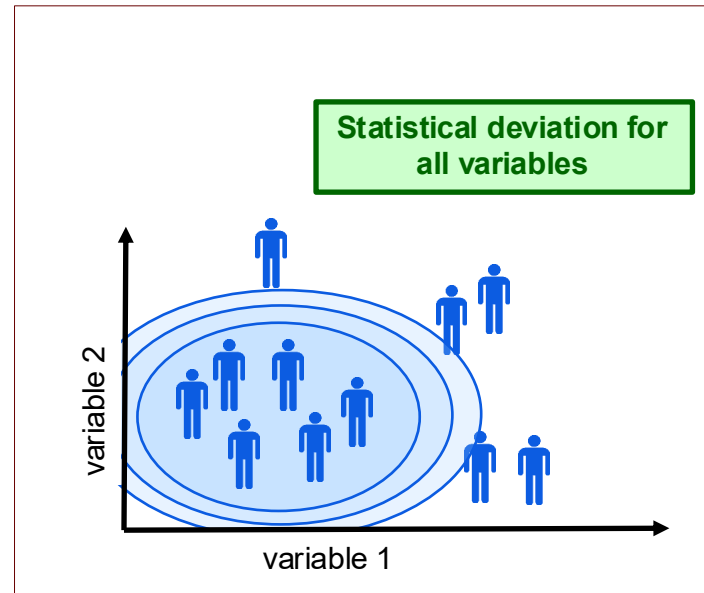
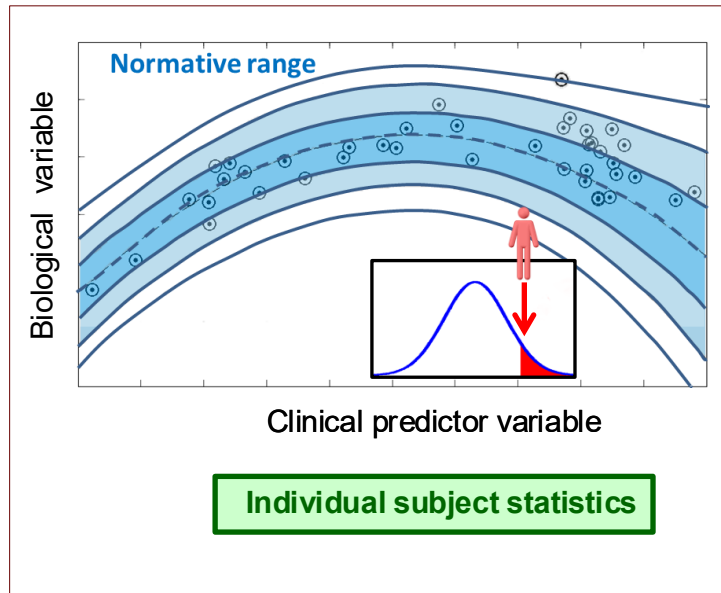


The models

Component of a normative model

Component	Example
Y: the variable we want to model	cortical thickness
X: covariate: something that we want to describe the outcome by	Age, Sex
X: Batch effects	Sex, field strength, study
F(x) A function to model the covariate	Hierarchical Bayesian Regression
Data and fit	Train and test split 2 step procedure: - Train the model; make predictions for observations of interest

Getting the z-score



$$Z_{nd} = \frac{y_{nd} - \hat{y}_{nd}}{\sqrt{\sigma_d^2 + (\sigma_{d_*}^2)}}$$

Getting the z-score

$$Z_{nd} = \frac{y_{nd} - \hat{y}_{nd}}{\sqrt{\sigma_d^2 + (\sigma_{d_*}^2)}}$$

$$\hat{Y} = f(X, \theta) + \epsilon$$

- σ^2 is the estimated residual variance from the likelihood
- σ^{2*} is the additional variance from parameter uncertainty (the posterior predictive variance)

NORMATIVE MODELLING

Y

Outcome/Target

gene expression, brain
volume, body mass,
metabolite concentration,
cell count...

$$Y = f(X, \theta) + \varepsilon$$

Outcome/Target
gene expression, brain
volume, body mass,
metabolite concentration,
cell count...

Covariates
(age, sex,
task parameters)
Batch effects

**Model
parameters**

Residuals

$$Y = f(X, \theta) + \varepsilon$$

Gaussian
process
regression



Warped
Bayesian
linear
regression



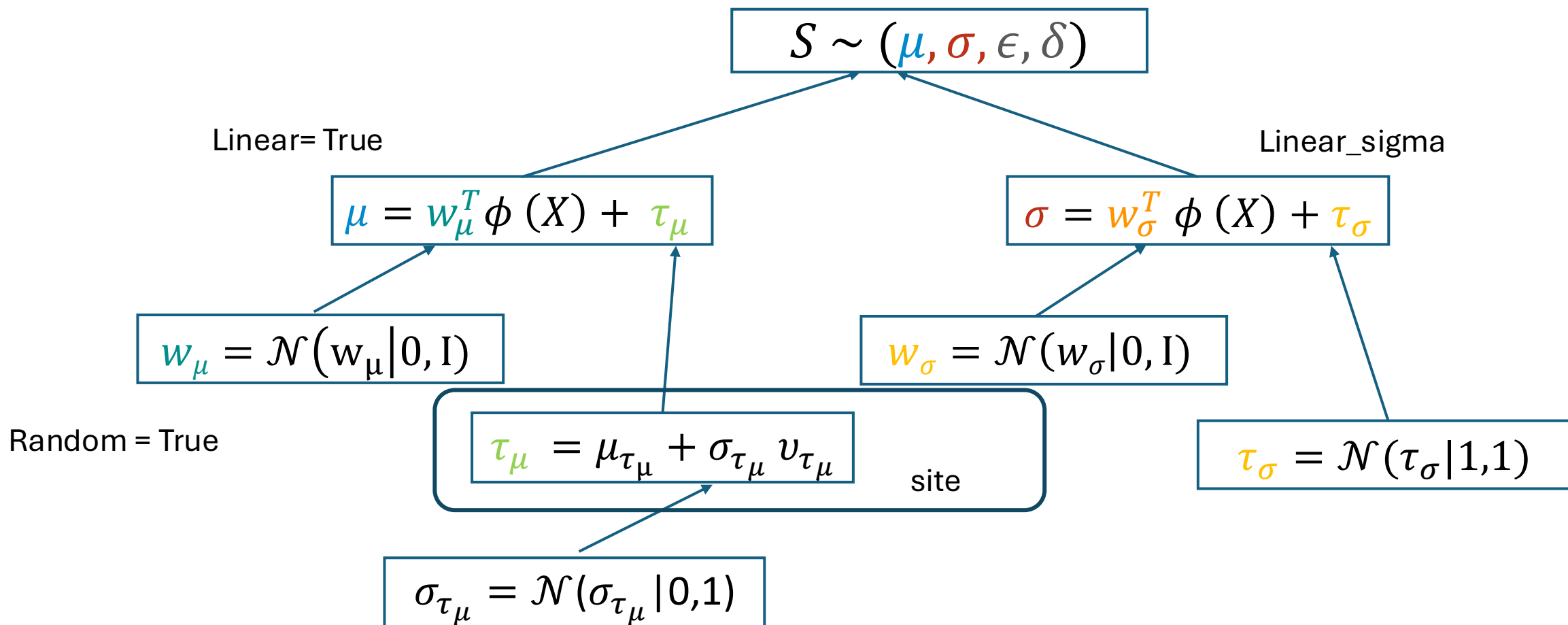
Generalized
additive models
of location scale
and shape



Hierarchical
Bayesian
regression



Hierarchical Bayesian Regression

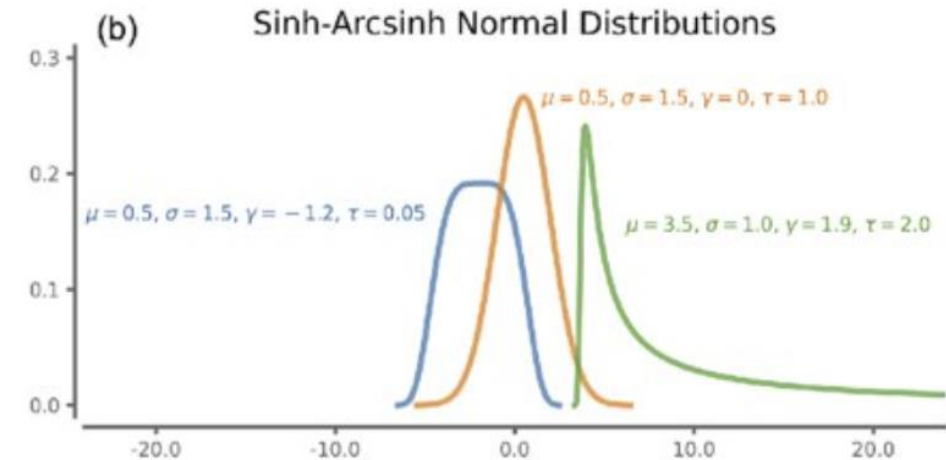


Spoiler: linear_delta and linear_epsilon for skewness and kurtosis also possible

SHASH Likelihood

$$S \sim (\mu, \sigma, \epsilon, \delta)$$

- Pymc: Nuts-sampler
- Reparameterisation of the SHASH (Sinh Arcsinh) distribution to make it suitable for an MCMC Bayesian sampling approach



Research Article



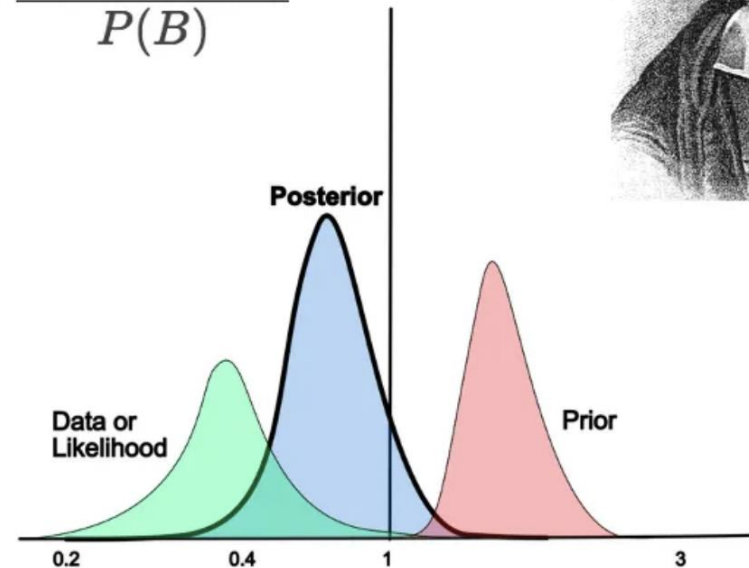
Non-Gaussian normative modelling with hierarchical Bayesian regression

Augustijn A. A. de Boer^{a,b}, Johanna M. M. Bayer^{a,b}, Seyed Mostafa Kia^{a,b,c}, Saige Rutherford^{a,b,d},
Mariam Zabihi^{a,b,e}, Charlotte Fraza^{a,b}, Pieter Barkema^{a,b}, Lars T. Westlye^{f,g,h}, Ole A. Andreassen^{g,h}, Max Hinne^a,
Christian F. Beckmann^{a,b,i}, Andre Marquand^{a,b,j}

The Bayesian Approach

- $P(A/B)$: posterior
 - $P(B/A)$: likelihood, data
 - $P(A)$ prior
-
- $P(B)$: marginal likelihood over B

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



Priors and hyperpriors

- Priors $P(A)$ and $P(B)$ are governed by priors themselves
- For example: Prior distribution over $P(A)$

$$P(A) = N(\mu, \sigma)$$

hyperparameters

$$\mu = N(0, 10)$$
$$\sigma = \text{softmax } N(0, 10)$$

hyperpriors

```
graph TD; HP[hyperparameters] --> PA[P(A) = N(mu, sigma)]; HPr[hyperpriors] --> mu[mu = N(0, 10)]; HPr --> sigma[sigma = softmax N(0, 10)];
```

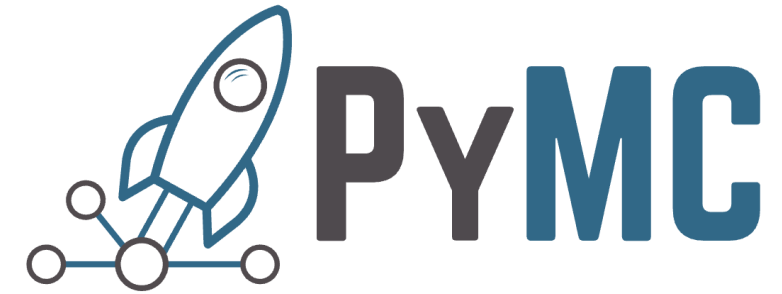
Shrinkage in Bayesian models

Pooling concept	Model behaviour	Effect type	Example
Complete pooling	Ignore groups	No group effect	$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$
No pooling	Separate parameter per group	Fixed effects	$y_i = \beta_{\{0,s[i]\}} + \beta_{\{1,s[i]\}} x_i + \epsilon_i$
Partial pooling	Share information across groups	Random effects	$y_i = \beta_{\{0,s[i]\}} + \beta_1 x_i + \epsilon_i$ $\sim (\mu_{\{\beta_0\}}, \sigma_{\{\beta_0\}})$

Bayesian modeling in science

- *Fully Bayesian:*
 - All parameters estimated jointly in one coherent inference
 - Uncertainty in hyperparameters propagates through to posterior
 - Priors, likelihoods, and hierarchical structure sampled together
- *Empirical Bayes:*
 - Hyperparameters estimated first (e.g. by maximising marginal likelihood), then fixed
 - Two-stage shortcut, faster but loses uncertainty propagation

PyMC



- Python Library for probabilistic programming
- Let's you define Bayesian Models in Python and then sample from their posterior using Markov Chain Monte Carlo
- User: describes the model iteratively
 - Priors, likelihood, hierarchical structure
- PyMC handles the sampling machinery
 - No-U-Turn Sampler, which is a variant of Hamiltonian Monte Carlo that is more effective than other methods like Metropolis Hastings

Why Do We need Sampling

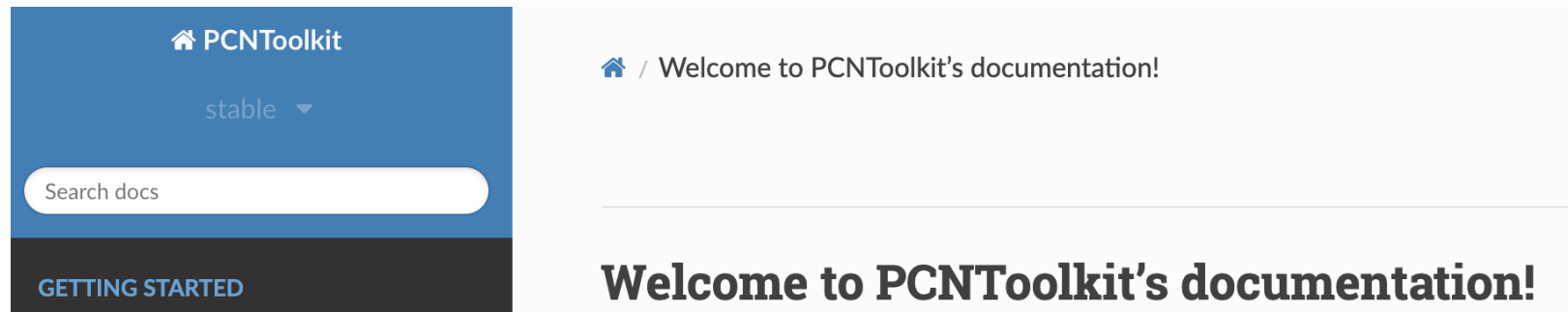
- The HBR posterior has no closed form, it cannot be calculated analytically. The posterior can be approximated by sampling
- MCMC approximates the posterior by drawing samples without needing the normalising constant
- With enough samples, expectations (centiles, Z-scores) can be estimated to any desired accuracy

Putting it all together:

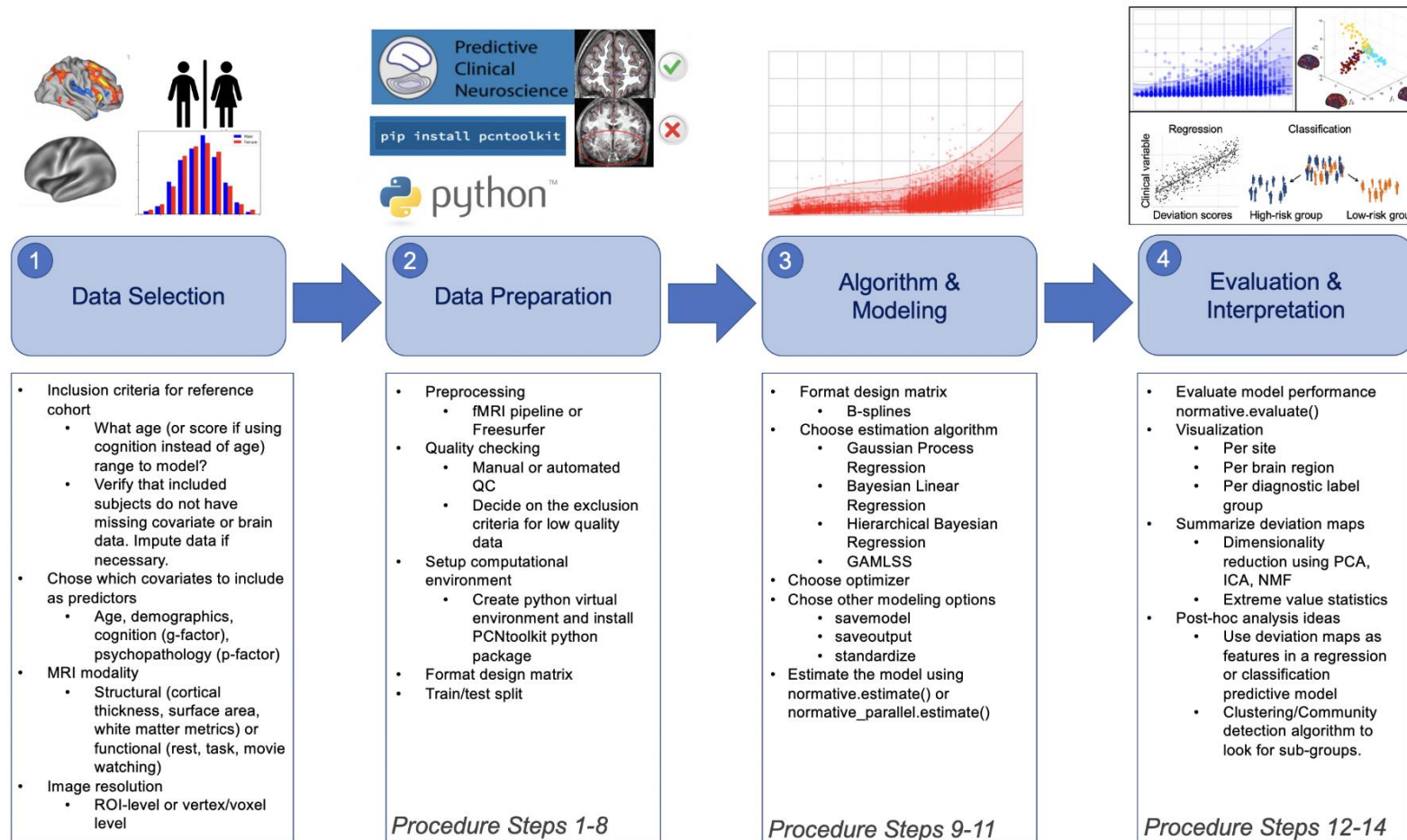
- Normative modelling is a "fancy" standardization technique that allows to express biological variance as a function of a chosen set of covariates and batch effects
- The model is implemented in a fully Bayesian approach

The PCNToolkit

- Object oriented Python Toolbox for normative modelling
- Based on PyMC and XDataArrays
- Comes with a set of predefined regression models
- <https://pcntoolkit.readthedocs.io/en/stable/>

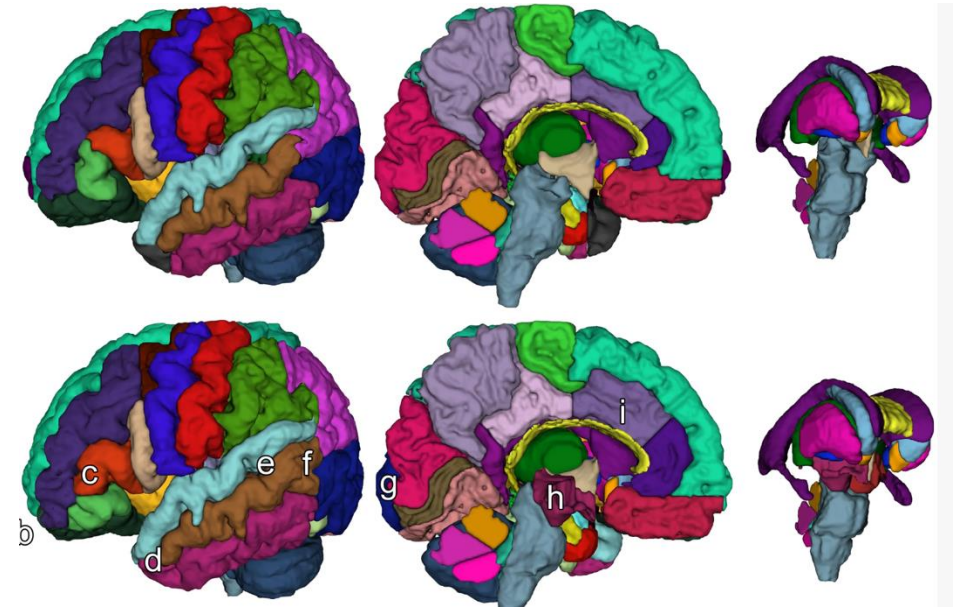


Steps involved



Data: fcon1000 data set

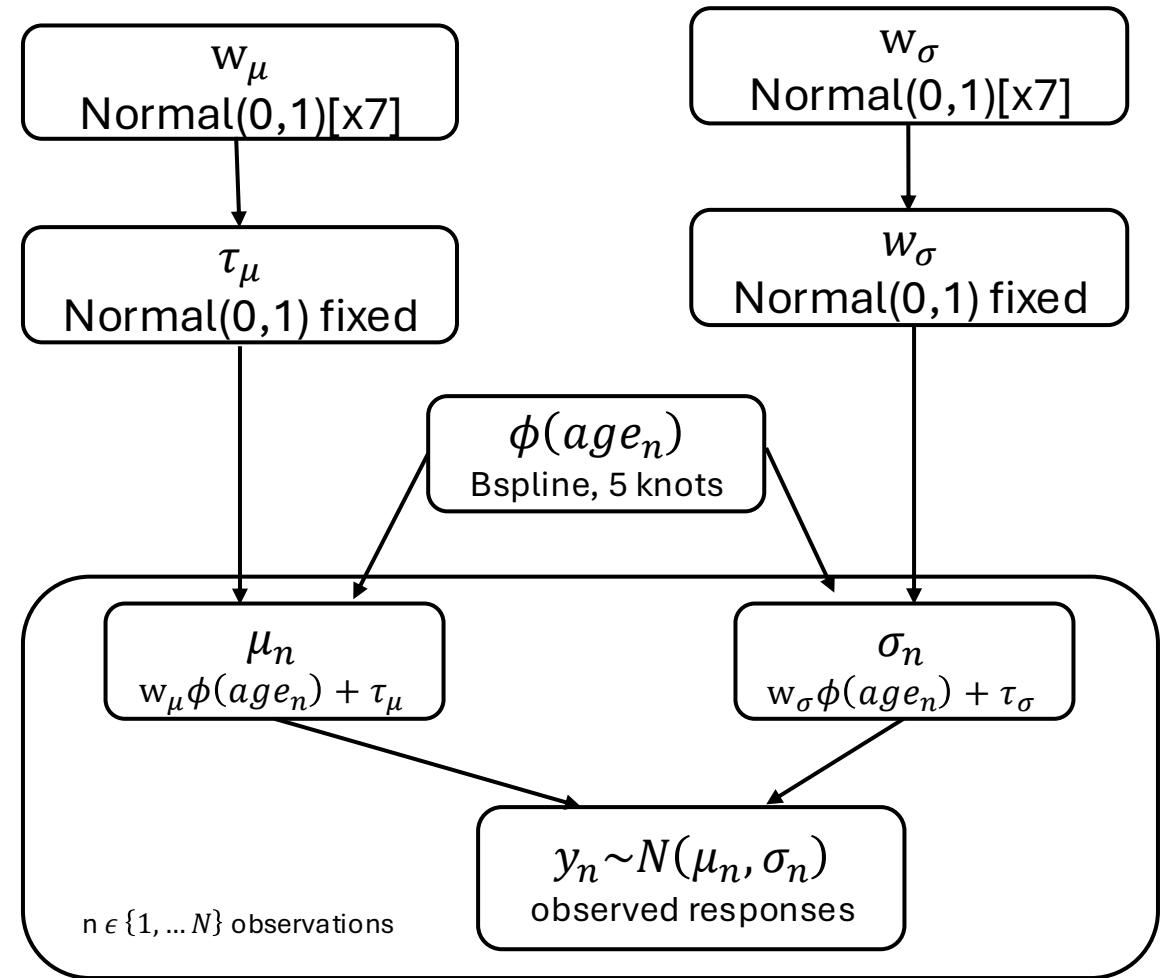
- The fcon1000 dataset is a publicly available multi-site neuroimaging dataset containing structural MRI measures from healthy control subjects
- Subset of the 1000 Functional Connectomes Project
- 1078 subjects
- Age 5-85 years
- Covariate age,
- Batch effects sex and site



<https://pubmed.ncbi.nlm.nih.gov/28479474/>

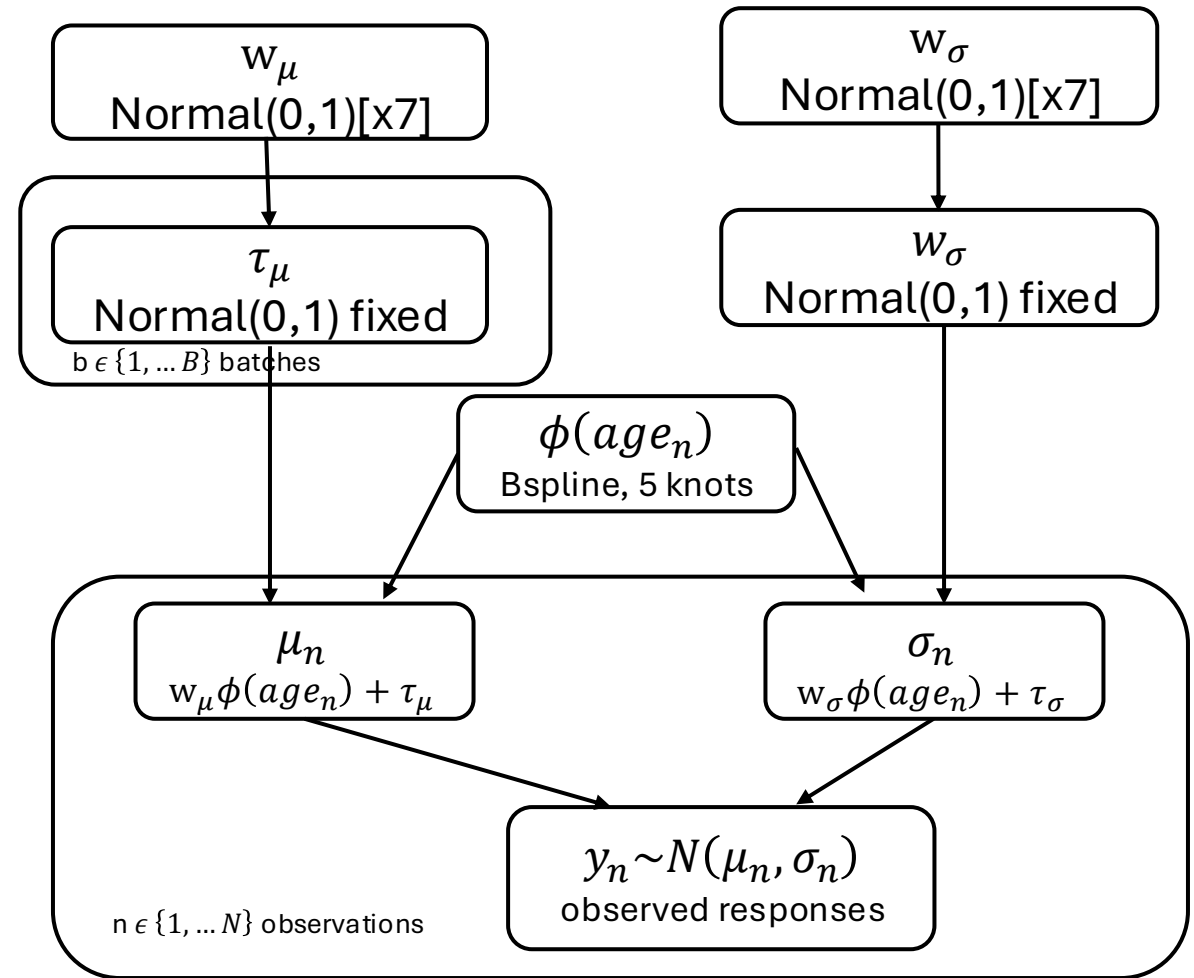
A simple model – no batch effects

- Gaussian Likelihood
- No batch plates
- μ_n and σ_n per observation
- τ_μ and τ_σ are single scalars shared across all sites and sexes



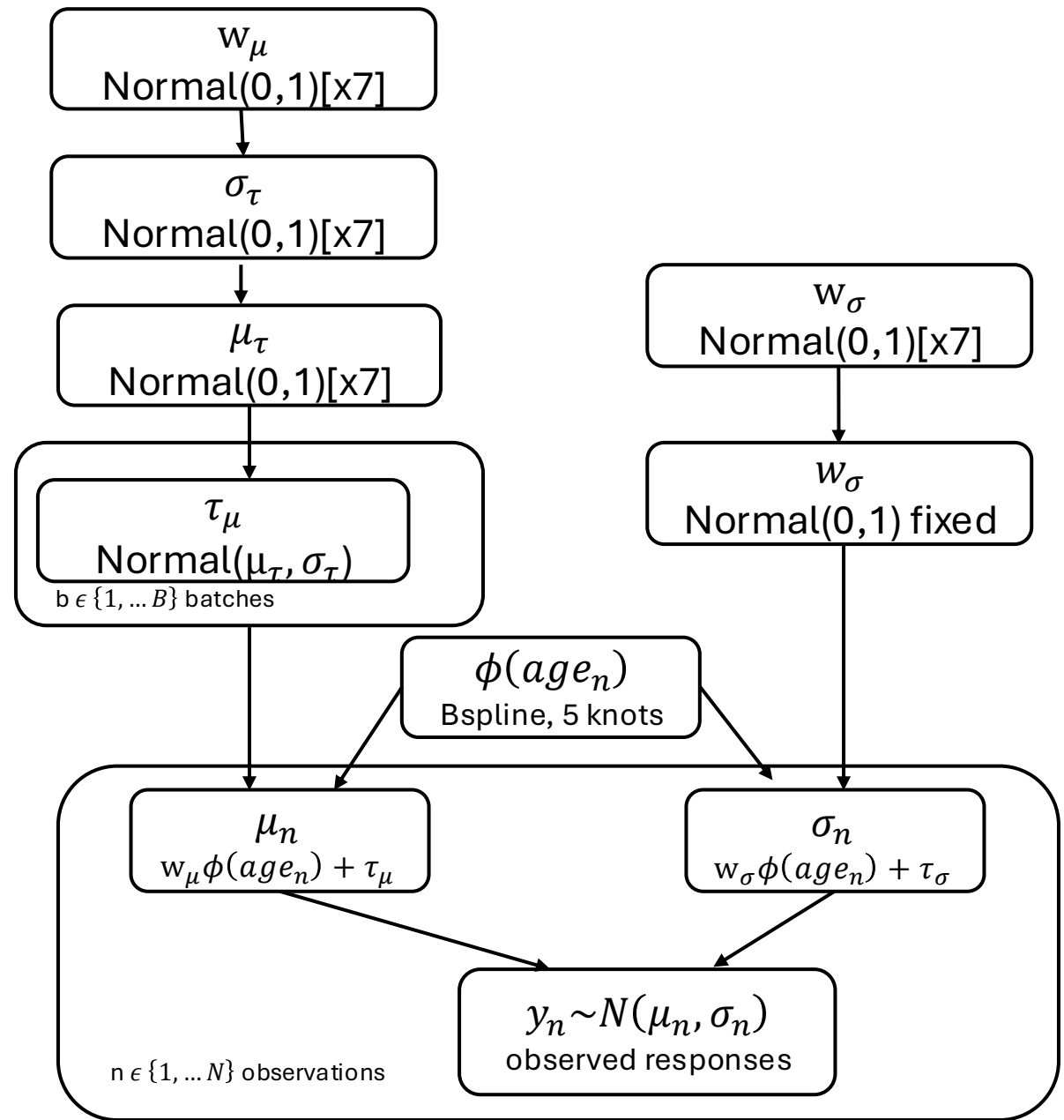
Fixed effects for site

- Each site x sex combination gets its own intercept τ_b , estimated independently from all other sites/sex combinations
- The age trajectory is still shared across everyone
- No hyperparameters above τ_b , a batch with 3 observations will be treated as a batch with 500



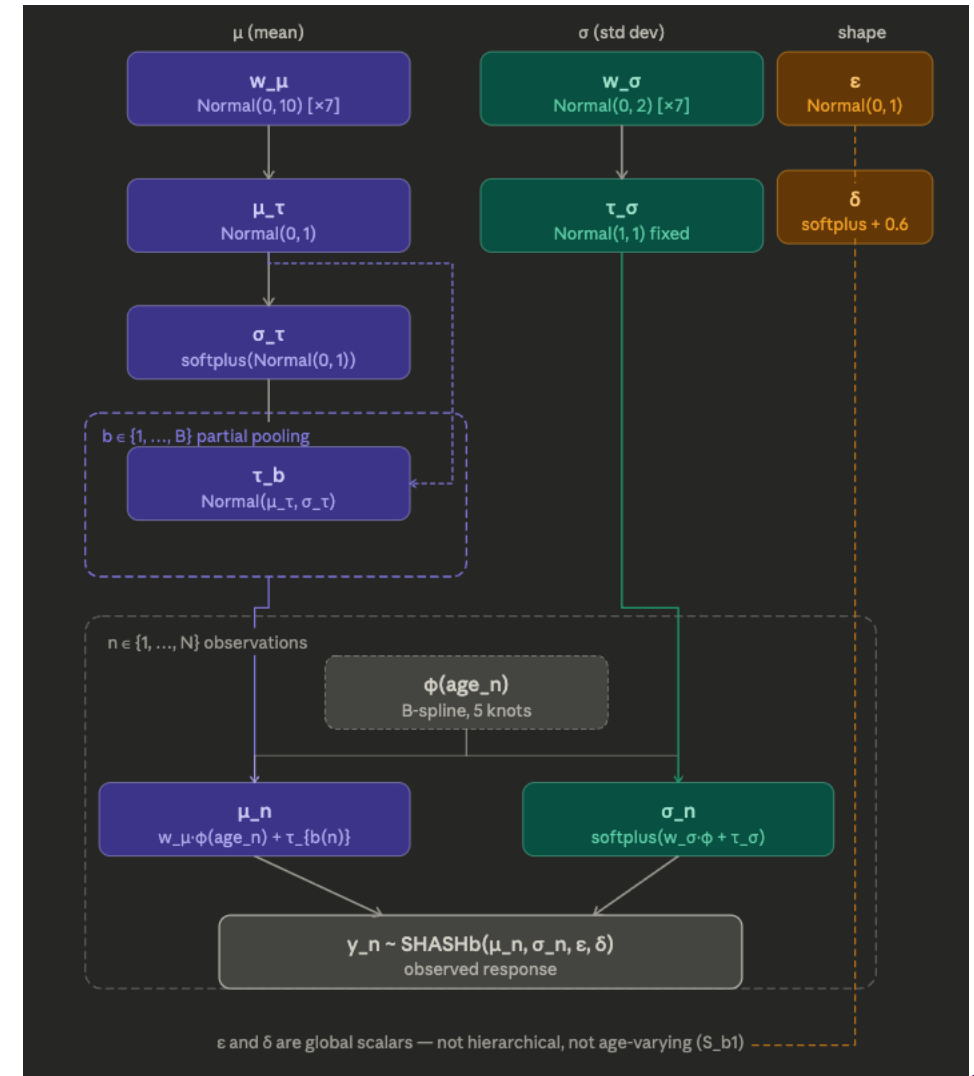
Partial Pooling Gaussian

- Site intercepts τ_b are drawn from a shared group distribution $N(\mu_\tau, \sigma_\tau)$, the hyperparameters are learned from all sites together
- Small sites are pulled toward the group mean (shrinkage); large sites deviate freely, the amount of regularisation is automatic and data-driven
- The model can generalise to new sites at test time by falling back on the group prior



Hierarchical Bayesian Regression

- Similar to last step, but including SHASH likelihood



Parameter	Role	Fixed intercept, fixed slope	Random intercept, fixed slope	Fixed intercept, random slope	Random intercept, random slope
μ	Mean of the distribution , the expected brain feature value for a given age, sex, site	One shared trajectory for everyone sites differ only if you add batch effects separately	Sites have different baseline brain feature levels but the same age trajectory shape	All sites share the same baseline but the rate of age-related change differs by site	Sites differ in both baseline and how the feature changes with age ,most flexible, most data-hungry
σ	Spread around the mean , biological variability between individuals of the same age	All sites have the same noise level and the same age-variance relationship	Sites differ in baseline noise level (some scanners are noisier) but variance changes with age the same way everywhere	Baseline noise is shared but the rate at which variance changes with age differs by site	Sites differ in both baseline noise and how variance evolves with age rarely justified
ϵ	Skewness: positive = right tail, negative = left tail	One global skew value for the whole dataset , the default S_b1	Sites differ in their baseline skew direction	Age-related change in skew differs by site but baseline skew is shared	Sites differ in both baseline skew and how skew changes with age , S_b2 territory
δ	Tail thickness : values below 1 mean heavier tails than Gaussian	One global kurtosis value for the whole dataset , the default S_b1	Sites differ in their baseline tail thickness	Age-related change in tail thickness differs by site	Sites differ in both baseline tail thickness and how tails evolve with age very flexible